

DIAGNOSE & FIX · YOUR HOME OR OFFICE

Fix Your Own Wi-Fi

A step-by-step, English guide to diagnosing your home
Wi-Fi connection with the free **WLAN Pros Toolbox**

ROAD ONE · WI-FI

ROAD TWO · INTERNET



For your phone

as you sit

The box on the wall

where the two roads meet

The wider world

your provider, then everywhere

BY

Parsons

wirelessLAN
PROFESSIONALS

Fix Your Own Wi-Fi

Stop guessing. In one tap, know whether it's your Wi-Fi, your internet, or just one bad app, and exactly what to do next.

Keith Parsons

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Chapter 1: “My Wi-Fi is broken” (and why it usually isn’t)



Most “my Wi-Fi is broken” problems aren’t Wi-Fi problems. That is the whole book in one sentence, and it is going to save you a lot of rebooting. The thing that froze your video call, stalled the kid’s game, or left a page spinning forever is real. It is just not always the part you think it is. By the end of this chapter you will tap one button on your device and know which part to look at first. Then you will know what to try, and what not to waste an afternoon on.

I have been designing, installing and troubleshooting Wi-Fi for over 25 years, and every single problem I have ever been called to fix started with the same four words. “My Wi-Fi is broken.” I have heard it from a CEO whose video call kept freezing. I have heard it from a hotel guest who could not pull up a boarding pass. I have heard it from my own grandkids when a game stalled on the far side of the house. Four words, said with total confidence, and so often wrong.

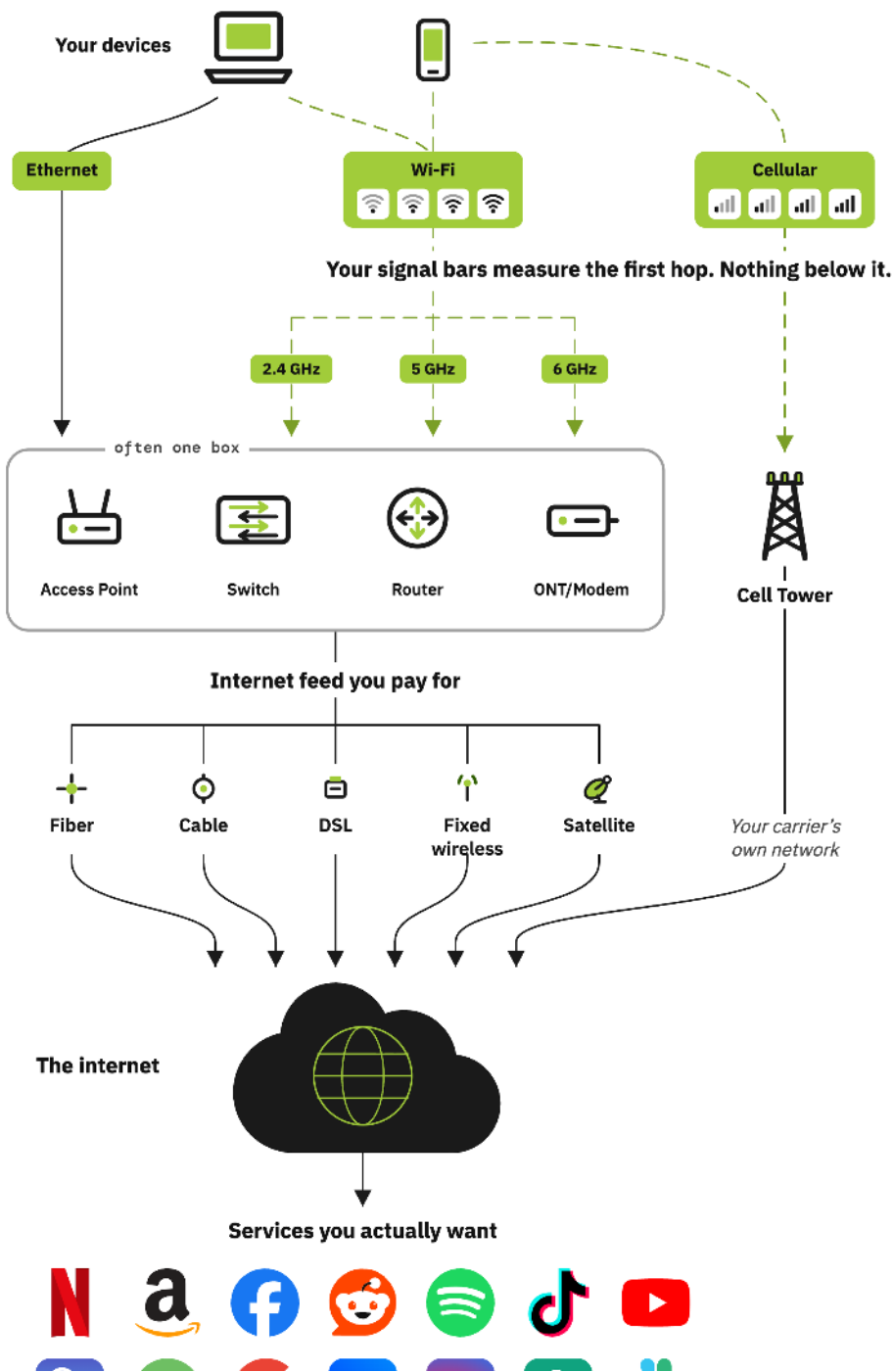
Here is what almost nobody stops to explain. Your Wi-Fi and your internet are two different things. They just happen to meet at one box on your wall, so they get blamed as one thing. The most common Wi-Fi complaint in the

world is actually an internet complaint wearing a Wi-Fi costume. Once you can tell the two apart, you stop guessing. So let's go look at the one screen that tells them apart for you.

The map, once, before anything else

Before you tap a single thing, look at this. It is the whole trip, start to finish, on one page.

How your devices access the Internet





The whole map. Your device on one side, the thing you actually wanted on the other, and everything in between. You do not need to understand all of it yet.

That picture is the map for this entire book, and you are not expected to take it in at once. Nobody does. What matters right now is that you have seen the shape of it, because from here on every drawing in this book is a close-up of one part of that same map. When a chapter needs to explain the short hop through the air, we will zoom in on the short hop through the air. When it needs to explain the part you pay your provider for, we will zoom in on that instead.

That is the whole method. One map, looked at closely, one piece at a time. So when a later picture shows you two boxes and an arrow, you will already know where those boxes sit in the bigger trip, and you will never be looking at something that came out of nowhere. Turn back to this page any time you lose your place. It is not going to change.

Before you start

This book uses the free WLAN Pros Toolbox. If you are reading this inside the Toolbox already, you have everything you need, so skip ahead. If you are holding the paperback or reading on a Kindle, take two minutes and install it now. It is free, there is no account to make, and nothing to buy.

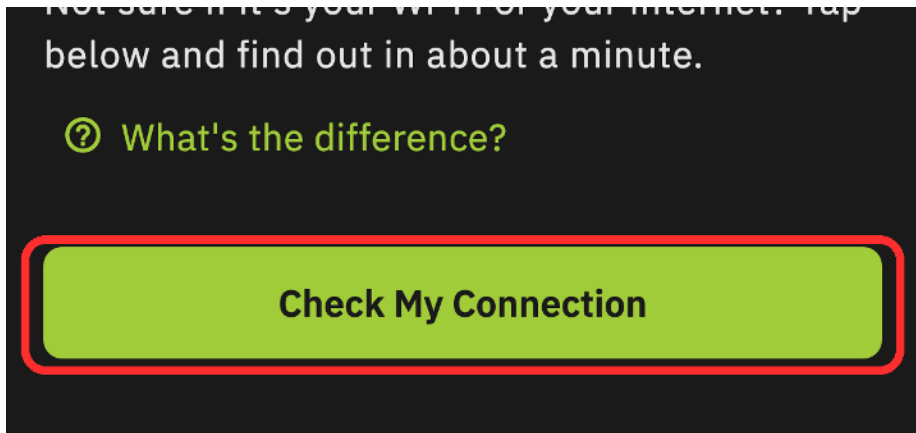
Go to toolbox.wlanpros.com on the device you want to test. That one page sends you to the right place for whatever you are holding: iPhone, iPad, Android, Mac or Windows. There is no version to pick and no edition to choose; there is one Toolbox and it is the same on all of them. On a phone it is the fastest route: type the address, tap the button for your phone, and you are done in under a minute.

If you would rather not type anything, the code on this page goes to the same address.

One thing worth knowing before you tap anything. What your device is willing to tell an app about your own Wi-Fi is different on an iPhone than on a Mac or an Android phone. That is Apple's rule, not ours, and the app works around it as best it can. If it ever cannot read something, it will say so

plainly rather than make a number up. You will see exactly that a few pages from now, and it is a feature, not a fault.

Now open the Toolbox and tap **Test My Connection**. Don't run anything yet. Just read the screen.



The Test My Connection screen, sitting quietly before you run it. One button to tap, the plain question it answers sitting above it, and a green link that explains the difference if you want it first.

Two roads, one box

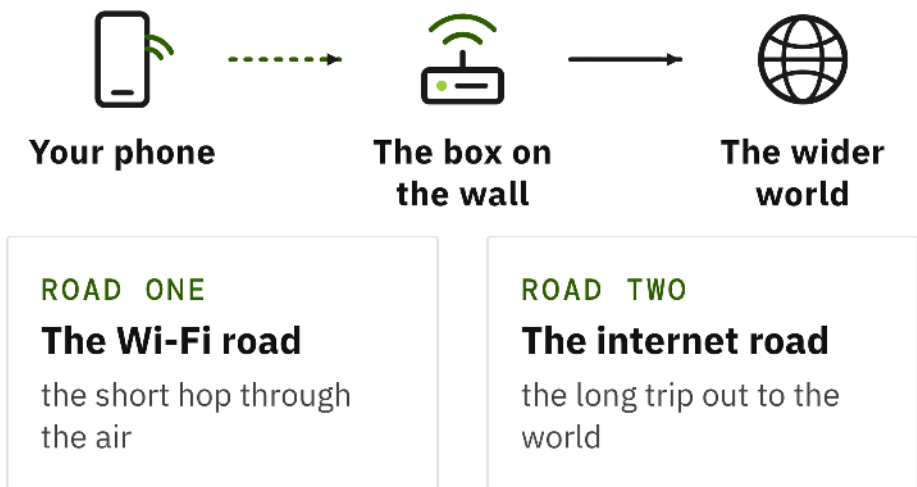
The screen is nearly empty, and that is on purpose. One button, **Check My Connection**, under one plain line:

Not sure if it's your Wi-Fi or your internet? Tap below and find out in about a minute.

That sentence is the idea most people never learn, sitting right there on your screen. Your Wi-Fi is one thing. Your internet is another thing. And this tool is about to grade them separately, one at a time.

Picture the trip your video call actually takes. It leaves your phone or laptop as a radio signal and floats across the air to a box on the wall. That hop through the air is the Wi-Fi road. From that box, your call runs down a cable, out to your internet provider, and across the world to whoever you are talking to. That long trip out to the world is the internet road. Two roads. One box where they meet. And when you are on Wi-Fi, the device in your hand starts its trip on that first road, the short hop through the air. That is the part it can see, and the only part it can tell you about directly.

One word, settled now so it never trips you up. **When I say the box, I mean whatever your device is actually talking to over the air:** the router your provider handed you, the gateway, a separate access point, or one of the little mesh units sitting around your house. If you have two or three of those small boxes on shelves in different rooms, that is a mesh, and it changes nothing in this book: your device is still talking to exactly one of them at a time, and that one is the box. Everything here works the same whether there is one box or four. The only difference is that with a mesh, the box nearest you may not be the one plugged into your provider's cable.



Two roads, one box. The short hop through the air is your Wi-Fi. The long trip out to the world is your internet. They meet at the box on your wall, which is why people blame them as if they were one thing. They aren't.

When the call freezes, the freeze could be anywhere on either road. But “my Wi-Fi is broken” already picked a side before anyone measured anything. That is the trap. You blamed the Wi-Fi road because it is the part you think is the problem. The network name you typed a password into once. The little bars up in the corner. The internet road is invisible to you, so it never gets blamed, even though it falls down much of the time.

There is an old line I keep coming back to, and it is deliberately provocative. *Wi-Fi usually works*. It just might not work the way you want. Far more often than not the radio hop is doing its job fine, and the slowness you are feeling is coming from somewhere else entirely. The trick is to stop guessing and let the device show you.

Run it, and read two answers, not one

Now tap **Check My Connection**. If your device asks permission the first time, allow it, and it will not ask again.

The tool does two jobs in one pass, and watching it work is the lesson. It reads your live Wi-Fi hop, the radio link to the box on the wall. At the same time, it reaches out across the internet road to a real spot on the far side of your provider and measures that too. Two readings, because there are two roads. It will not blur them into one happy number.

Watch the two results settle. Each one grades its own road: **Strong**, **Moderate**, or **Weak**. Above them sits a headline written in plain English, and it points at the side that deserves your attention first. That headline is the answer to “my Wi-Fi is broken.” Read it first.

The screen says a little more than which road is weaker. In plain words, it tells you about how much faster or slower one road is than the other, so you feel the size of the gap without doing any math. It also invites you to walk a few steps and run it again, because the answer moves as you move. Read that as a promise: the next chapter puts both roads on one screen and lets you watch them swing apart with your own feet.

There are four ways this can land, and each one tells you where to start.

The Wi-Fi result is the weaker grade. The radio hop is the weak link right now. Your internet feed is the healthier of the two. Something between you and the box on the wall is struggling. Good news: this is often a problem you can fix yourself, usually by getting closer or moving the box, and the next few chapters teach you how.

Your Wi-Fi is the slow part.

🔄 Run again

Wi-Fi:

❗ Weak

Internet:

✅ Strong

Wi-Fi the weaker grade. Wi-Fi reads Weak, the internet reads Strong, and the headline points at the radio hop, the part you can usually improve yourself.

The Internet result is the weaker grade. Now it is flipped. Your radio hop looks fine, and whatever is holding you back sits further out, somewhere past your own four walls. Standing closer to the box will do exactly nothing here, so stop dragging furniture around. Run it once more to be sure the reading holds. If it does, the next place to look is the equipment where that feed arrives, and after that, your provider. We will walk through what to check next, and when it is time to make the call.

Your internet is the slow part.

🔄 Run again

Wi-Fi:



Strong

Internet:



Weak

Internet the weaker grade. No amount of standing near the box fixes this one. Confirm the reading, then start working outward from where the feed comes in.

Both results read Strong. The screen says so in plain words: *Both sides are strong. They're about the same speed.* Both roads look healthy, which means the thing that is actually slow is most likely the one app or website you were using. The screen tells you to look past your connection in the lines below that headline. This is the answer people refuse to believe, and it is the most useful one in the book.

**Both sides are
strong. They're
about the same
speed.**

 **Run again**

Wi-Fi:  **Strong**

Both roads strong. The headline reads “Both sides are strong. They’re about the same speed.” Your connection did its job. Try a different app or a different site before you change anything at home.

A result says “Couldn’t check.” Sometimes the tool reads one road but not the other. When that happens it tells you so, honestly, instead of inventing a number. An honest “I couldn’t see that” beats a confident wrong answer every time. Tap **Run again**. If it reads the second time, carry on. If it still cannot see one side, don’t treat that missing answer as a bad answer. It just means you need a little more information before you decide anything.

We checked your internet, but not your Wi-Fi.

↻ Run again

Wi-Fi:

❓ Couldn't check

Internet:

⚠️ Moderate

“Couldn’t check” is not the same as Weak. The tool will never hand you a fake number. Tap Run again, and if it still cannot see that side, go get more information rather than guessing.

“Both strong” is the answer you need most

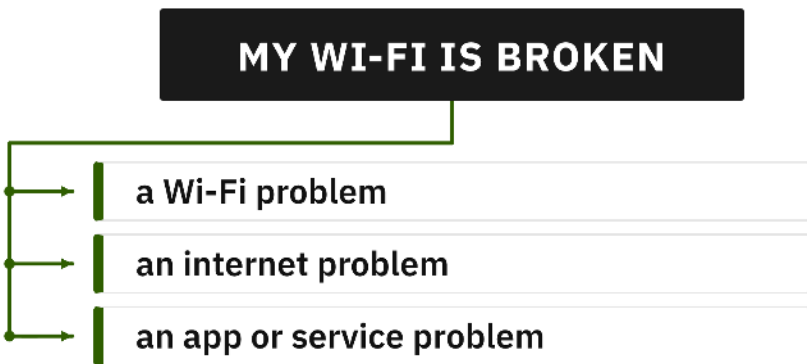
The answer that frustrates people the most is the one that helps them the most. Both results read Strong. They wanted a villain, and the tool just told them everything looks healthy.

Here is what that answer is really saying. Your Wi-Fi delivered. Your internet delivered. So before you change anything at home, prove it. Try another app, or another website. If nine things work and one does not, start by suspecting the one. What is slowing you down is most likely somewhere this tool cannot reach and you do not own: one overloaded website, a video service choking during a busy hour, an app having a bad afternoon on its own end. Your connection is not where to start looking.

That is a gift, because it stops you from doing the three useless things people do when Wi-Fi “feels slow.” They reboot the box that was working fine. They go stand right next to it for no reason. They call the provider and wait on

hold to be told nothing is wrong. “Both strong” hands you the move that actually helps: open a different app, open a different website, or wait twenty minutes and try again. If nine things work and one does not, start by suspecting the one.

There is one more reading worth knowing. The two results can also land low together, both Moderate or both Weak at once. That is a different message. “Both strong” means look past your connection. Both low means start with the Wi-Fi hop, and here is the reason, because it is worth understanding rather than memorizing. The internet reading is measured *through* your Wi-Fi. Everything it sends and receives crosses that same short hop first. So a weak hop drags the internet number down with it, and the two readings sink together even when the feed coming into your house is perfectly healthy. A low internet number next to a low Wi-Fi number is not proof that both roads are bad. It is exactly what one bad road looks like from here. So fix the near end first. Move closer to the box and run it again. If the Wi-Fi side improves when you move, so will the other one, and you have found your answer. If both stay low standing right next to the box, then the hop is not the limit and the feed is worth a harder look.



One complaint, three real causes. The Wi-Fi hop, the internet feed, or one app or site. Test My Connection is the fork that tells you which path to walk down first.

Back to those four words

Return to “my Wi-Fi is broken.” You now have a button that answers it in one pass.

That video call that kept freezing? Run Test My Connection during the next freeze. If the Wi-Fi result is weaker, you have a radio hop to chase, and you

can usually fix it. If the Internet result is weaker, stop trying to improve the Wi-Fi and start working outward from where the feed comes into your home. If both read Strong, leave both roads alone and go try another app, which will save you an afternoon of rebooting the wrong thing.

Notice what just happened. You did not learn one single fact about radios or signal or channels. You learned that the complaint blames one road when the trouble might be on the other, and you learned to read two answers where everyone else reads one. That alone puts you ahead of nearly everyone who has ever said “my Wi-Fi is broken.”

What this chapter can't tell you yet is the *why*. When the Wi-Fi result grades low, what is your radio actually doing? And when something is slow, slow how? Slow in the way that wrecks a video call, or slow in the way that only annoys a big download? Those are different problems with different fixes, and the next chapters take them apart one at a time. You have the one-tap verdict now. Next we slow it down and look at why the two roads move on their own.

Do this now

Run Test My Connection three times, in three different spots. The point is to feel, in your own hand, that the two roads move on their own.

1. **Run it once where you are sitting right now.** Note the grade on each result. Read the headline and decide whether you agree with it before you read on.
2. **Walk to the farthest corner of your home and run it again.** Watch the Wi-Fi result. If it slips toward Weak while the Internet result holds steady, you just watched the radio road weaken while the internet road stayed exactly the same. Two roads, moving on their own.
3. **Run it one more time right next to the box on the wall.** The Wi-Fi result will usually be at its strongest here. If the Internet result is still the weaker one, no amount of standing close fixes it, and now you have the proof in your hand.

Three runs, and you will never again say “my Wi-Fi is broken” without first asking which road actually needs the attention.

Myth-Bust: “Full bars means the internet is working.”

No. The bars in your corner only ever describe the first road, the short radio hop to the box on the wall. They say nothing about the internet road at all. You can sit under five strong bars with a dead internet feed. You can have a perfect internet feed sitting behind one weak bar. The bars and the internet are two separate facts about two separate roads, and the bars only ever spoke for the first one. So stop reading “full bars” as “everything’s fine.” It was never saying that. And here is the friendliest version of the same idea: if a Wi-Fi network ever asked you to sign in on a web page, type your room number, or accept the terms, your Wi-Fi carried that page to you. The radio hop was working. Look at the other road first.

Note: Two kinds of bars, two different roads.

Look at the top of your screen and you will usually find two little stacks of bars, and they do not mean the same thing. The Wi-Fi bars, the little fan shape, only measure the short hop from your device to the box on your wall. From that box your traffic still has to travel out through a cable to your provider before it reaches the internet, and the Wi-Fi bars know nothing about that part. The cellular bars, the climbing stair-step, measure something else entirely: the link from your device to your carrier’s nearest cell tower, which is its own separate road to the internet that never touches your box at all. Full Wi-Fi bars sitting on a dead internet feed is an everyday thing, because the Wi-Fi bars were only ever grading the hop to the box. And here is the part people miss in a dead spot at home: if your Wi-Fi bars are full but nothing loads, switching that one device off Wi-Fi and onto cellular sends it out the other road entirely, the tower instead of your box, and often everything starts working again. Same internet, a completely different road to reach it.

Next: Chapter 2 takes the two roads apart on their own screen, so you can watch your Wi-Fi and your internet move independently and finally understand why the bars in the corner lie.

Chapter 2: The two roads (Wi-Fi vs the internet, made obvious)

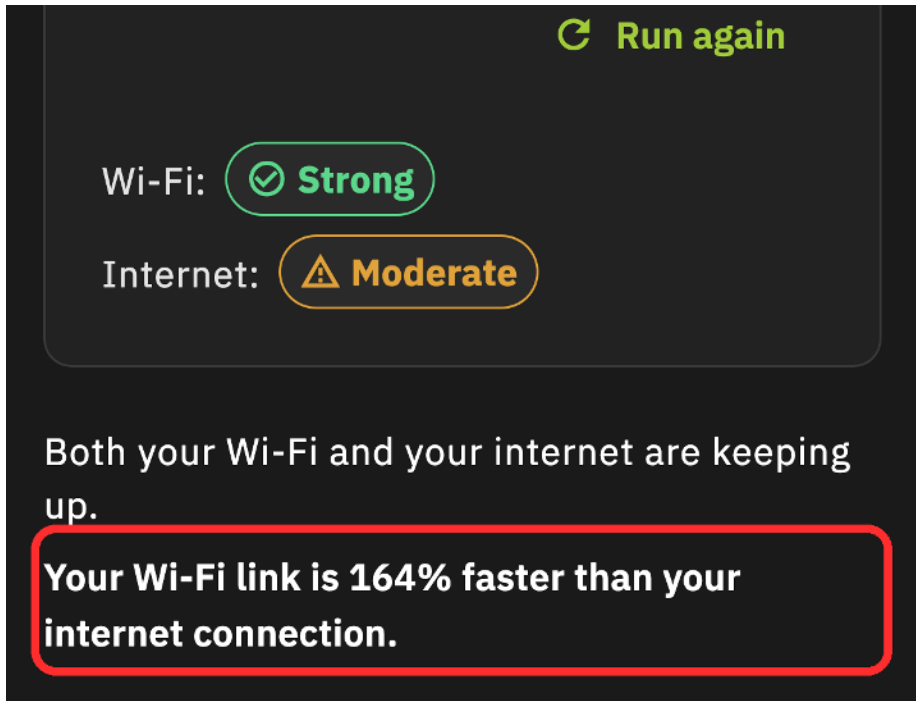


A faster internet plan does not make your Wi-Fi faster. That sentence is going to save you money, and it is the whole point of this chapter. The check in the last chapter handed you two grades because there are two separate things to grade, and most people have never once seen them apart. By the end of this chapter you will have watched your Wi-Fi and your internet move on their own, on one screen, and you will understand for good why the bars in the corner of your phone keep lying to you.

Here is the trap, stated plainly. Your home has two roads that carry everything you do online, and they meet at one box on your wall. The first road is the short hop through the air, from your phone to that box. That is your Wi-Fi. The second road is the long trip out from the box to your provider and the rest of the world. That is your internet. Your plan buys capacity on the second road. It does not make the first road stronger. So when someone sells you a bigger plan, they are widening the second road and leaving the first one exactly as it was. If the slow part was the first road, you just paid more money to fix the wrong thing.

If you have ever upgraded your plan and the back bedroom was still slow, you already felt this in your own home. You did the obvious thing, you spent the money, and the problem sat there grinning at you. Now you are going to see why on one screen, instead of guessing.

Open the WLAN Pros Toolbox on your device. Tap **Wi-Fi vs Internet**. This is the screen built to put the two roads side by side and prove they move on their own.



The Wi-Fi vs Internet screen. One plain verdict line that names which road is faster and by how much. This is the screen that proves the two roads are two different things.

Two roads, side by side

Run it, and read the one line at the top. It says something like *your Wi-Fi is faster than your internet* or *your Wi-Fi is slower than your internet*, with a rough sense of by how much. That single line is doing something almost no one has ever seen. It is reading both roads at once and telling you which one is the tighter of the two right now.

It is worth knowing exactly how it gets those two numbers, because this book keeps asking you to trust measurements and you should know what you are trusting. **The internet side is measured:** the app actually pulls data across that road and times it. **The Wi-Fi side is worked out rather than raced:** your device and the box on the wall negotiate a rate between them, the app reads that rate, and then it takes a little over half of it. Radio never delivers everything it advertises. Roughly half to sixty percent of the negotiated rate survives as real throughput, and the app uses that instead of quoting you a number you will never actually see. The screen says so itself in a footnote, which is the whole point. **So the comparison is honest: a realistic estimate of what your Wi-Fi hop could carry, against a real measurement of what your internet feed just did carry.** It is not a stopwatch race between the two, and it does not claim to be.

Think about what that means for a second. The tool just compared the short hop through the air to the long trip out to the world, and one of them came out smaller. For anything that needs room to move, a big download, a video stream, a house full of people watching different things, the smaller road sets the ceiling. That is about how much fits, not about every kind of slow. There are three other kinds, and Chapter 4 is where they live. The bigger road does not rescue it. A wide internet road feeding a narrow Wi-Fi road gives you narrow-road speed. A wide Wi-Fi road fed by a narrow internet road gives you narrow-road speed the other way. Traffic can only move as fast as the tightest part of the trip.

This is where the plan upgrade falls apart. The plan only ever touches the internet road. If the verdict says your Wi-Fi is the slower of the two, a bigger plan widens a road that was never the problem. You can pay for the fastest internet plan your provider sells, walk to the back bedroom, and crawl, because back there the Wi-Fi road narrowed to a country lane and no amount of internet plan touches it.

Watch them move

Now walk around your home and run it again in a few different spots. Watch the verdict flip.

Stand right next to the box on the wall and run it. Out here, close in, your Wi-Fi road is usually wide open, so the verdict often says your Wi-Fi is the

faster of the two. The internet road is the limit, because the Wi-Fi hop has nothing in its way. Now carry the phone to the farthest corner of your home, behind a wall or two, and run it again. Watch the verdict swing the other way. Out there the Wi-Fi road narrows, sometimes a lot, while the internet road did not budge an inch. Same provider, same plan, same box, and the answer changed completely because you moved fifteen steps.

That little experiment is the entire chapter in your hand. Same provider, same plan, same box, and the answer changed because you moved. The internet road held steady while the Wi-Fi road changed under your feet. Two roads. One box. Moving on their own.

A hotel, and a hard lesson

I learned this one the hard way in a hotel, and the story makes the point better than any diagram.

The Wi-Fi in this place was beautiful. Strong everywhere, no dead spots, the kind of signal that lights up every bar on every phone. By every sign a normal person reads, this was great Wi-Fi. And it was useless. A web page took forever. A video would not start. Everyone in the building was sure the Wi-Fi was broken, because the Wi-Fi is the part you can see, and the part you can see looked perfect.

It was not the Wi-Fi. The short hop through the air, from every phone to every box, was doing its job beautifully. The problem was the other road, the long trip out to the world, the feed the hotel was paying for. That road was a trickle. The whole building was trying to push everything it did through a pipe the size of a coffee stirrer. They had fantastic Wi-Fi. They had a terrible internet feed. The first road was wide open and the second road was nearly closed, and because everyone only ever looks at the first road, everyone blamed the wrong one.

Here is why that story matters in your own home. If that hotel had walked the building adding more Wi-Fi, buying stronger boxes, chasing fuller bars, they would have spent a fortune widening a road that was already wide open. The slow part was the road nobody was looking at. Your home is the same shape, just smaller. Before you spend a dollar widening either road, you want to know which one is actually narrow. The Wi-Fi vs Internet screen tells you in one line.

Back to the bigger plan

Return to the thing that started this chapter. Someone is about to sell you a faster internet plan to fix your slow Wi-Fi.

Run Wi-Fi vs Internet first, in the spot where it actually feels slow. If the verdict says your internet is the slower road, you have learned where to look, and that is not the same as learning what to buy. A slow internet side can mean the plan is too small, and it can just as easily mean congestion, an outage, tired equipment, or a bad day on a fixed-wireless or satellite link. So before you spend anything, find out whether you are getting what you already pay for. But if the verdict says your Wi-Fi is the slower road, put your wallet away. A bigger plan widens the road that was already fine and leaves the narrow one exactly as narrow as it was. You would be paying every month, forever, to fix a problem you do not have.

WILL FASTER INTERNET FIX THIS?

Wi-Fi is the weaker road → **No. Fix the Wi-Fi first.**

Internet is the weaker road → **Check what you already pay for before buying more.**

Both look healthy → **Don't buy anything yet.**

Will faster internet fix this? Three answers, and the screen tells you which one is yours. If the Wi-Fi side is the weaker road, a bigger plan buys you nothing. If the internet side is the weaker one, the first question is not what to buy, it is whether you are getting what you already pay for. And if both look healthy, the trouble is somewhere else entirely and your wallet stays shut.

Notice what you can do now that you could not do at the start of the chapter. You can stand in the slow spot, run one screen, and know whether to start with the feed coming into your home or with a few steps closer to the box. You can tell the difference between a road you pay for and a road you do not. That alone puts you ahead of almost everyone who has ever upgraded a plan in frustration and gotten nothing for it.

What this chapter cannot tell you yet is what is happening on that first road when it goes narrow. When your Wi-Fi is the slower one, what is your connection actually doing back there in the corner? Your device already knows, and it will show you five plain facts that explain the whole thing. That is the next chapter. You have the two roads now. Next we walk up to the first one and read it.

Do this now

One reading, in the spot that already annoys you, and one question worth money.

You have already walked your home with Chapter 1 and watched the two roads move apart. You do not need to do that again. This time you are standing still and asking the screen a different question: which of the two roads is the narrow one right here, and what would money actually fix?

1. **Go to the spot in your home where things feel slow.** Not a fair test spot, not next to the box. The chair where the video stutters, the bedroom where the page hangs. That is the only spot whose answer matters, because that is where you live.
2. **Run Wi-Fi vs Internet once and read which road is the narrower one.** That single line is the whole answer. One of your two roads is the limit in this chair, and the screen just named it.
3. **Now spend the money in your head.** If the screen said your **internet** is the narrower road, that is where to investigate, starting with whether you are getting the service you already pay for, not with buying more of it. If it said your **Wi-Fi** is the narrower road, stop. A bigger plan widens a road that was never the problem, and you would pay every month for a change you cannot feel in this chair.

One reading, and you know whether the next upgrade you are sold is aimed at your actual problem or at the other road entirely.

Myth-Bust: “A faster internet plan means faster Wi-Fi.”

No. A plan only ever touches one of your two roads, the long trip out to your provider and the world. It never touches the short hop through the air from your phone to the box on the wall, and that hop is the part that goes narrow in the back bedroom. You can pay for the fastest internet plan sold in your town and still crawl in the far corner of your own home, because back there it was the Wi-Fi road that narrowed, and the Wi-Fi road is not for sale. Whatever feels slow, run the check before you reach for your wallet. If your Wi-Fi is the slower road, a bigger plan is money handed over to fix a problem you do not have.

Note: Cellular does not use either road.

Everything in this chapter has been about two roads that meet at the box on your wall. Your phone has a third way out that skips both of them. When it uses cellular, it talks straight to your carrier’s nearest tower and reaches the internet from there, never touching your box, your Wi-Fi, or the feed you pay your provider for. So when the Wi-Fi vs Internet screen tells you which of your two roads is the tighter one, it is telling you nothing at all about the cellular road, because that road was never part of the comparison. The two meters at the top of your screen are not the same meter, and they are easy to tell apart once you look. The curved arcs, the little fan, are Wi-Fi, and they only ever describe the hop from your device to the box. The climbing stair-step bars are cellular, and they only ever describe the hop to the tower. Neither one has any idea what happens after its own first hop, and neither one can speak for the other.

Here is the part that catches people at home. Your phone usually has both radios available at once. When you are on Wi-Fi, nearly everything goes over Wi-Fi, but a phone can quietly lean on the tower when the Wi-Fi is struggling. That means you can run this chapter’s test, read a perfectly good verdict about your two roads, and the thing that felt slow may have been going out over the tower instead. It also means you have a free experiment available whenever a corner of your home goes bad. Turn Wi-Fi off on that one device, watch it fall back to cellular, and see whether the trouble follows it. If everything suddenly works, the problem was on one of the two roads in this chapter. If it fails over cellular too, you have learned something just as useful and far more specific: your home network is not the suspect. Start with the device, the app, or the service it is trying to reach.



Your signal bars measure the first hop. Nothing below it.

The two meters side by side, zoomed out of the big picture. The fan on one side, the stair-step bars on the other, and the line underneath that settles it: they are grading two different first hops, and neither one can speak for the other. Same drawing as Chapter 1, camera pushed in on the pair of them.

Next: Chapter 3 walks right up to that first road and reads it. Your own device already knows five plain facts about your Wi-Fi connection, and once you can read them, a slow corner stops being a mystery.

Chapter 3: Reading your own connection (the five facts)



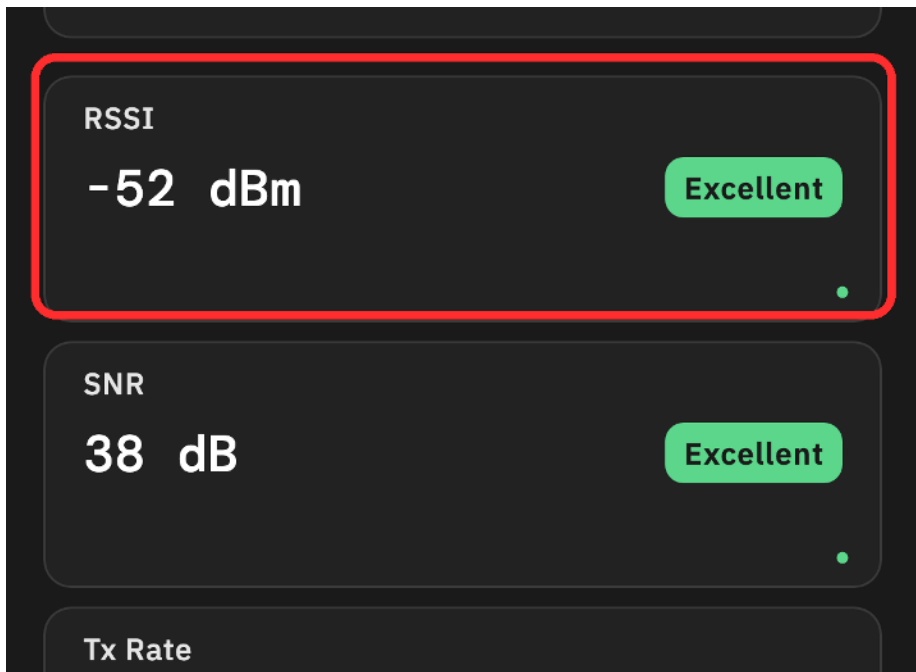
Your own device already knows most of what you need to know about your Wi-Fi, and it will tell you in five plain words. That is the whole chapter, and it is going to turn a slow corner from a mystery into something you can read. You do not have to learn radios. You do not have to memorize a single number. You tap one button, and your connection lays out five facts in plain language, and each one is either fine or it is the problem. By the end of this chapter you will read a good connection and a bad one side by side and know, at a glance, the shape of each.

The last chapter handed you two roads and proved your Wi-Fi is the one you can usually fix yourself. This chapter walks right up to that road and reads it.

Go back to the map for a moment. On the big picture from Chapter 1, everything in this chapter happens in one place: the very first leg, your device to the box on the wall. That is the only leg of the whole journey with a meter on it. Everything downstream, the feed out to your provider and the long haul to the world, runs with no gauge you can read at all. So the five facts in this chapter are not five facts about the internet. They are five facts

about one short hop, and the reason they matter so much is that it is the only hop you can see and the only one you can move. When your Wi-Fi is the slower of the two, these five clues explain most of what a home needs to know about it, and your device already measured all five. It just never showed them to you in a way a normal person could use. The Wi-Fi Information screen does exactly that. Five facts, five plain meanings, no math.

Open the WLAN Pros Toolbox on your device. Tap **Wi-Fi Information**. On an iPhone it asks once for permission and then reads your live connection in a single tap. Read the five things it shows you. We will take them one at a time.



A good connection. The right network, a fast lane, room to breathe, an Excellent signal, and a healthy link speed. This is the shape you want, and once you have seen it, the bad one is obvious.

The five facts your device already knows

Five things decide whether your Wi-Fi works. Your device reports all five, and here is what each one means in plain language, with no number you have to remember.

Which network. This is the name of the Wi-Fi you are actually connected to right now. It sounds too simple to matter, and it is the single most common surprise in this whole book. Devices grab onto a network on their own, and they often grab the wrong one: the slow guest network, the old network name you stopped using, the neighbor's signal you joined once on a visit, the phone hotspot still sitting in range. Before you chase anything else, read the name and make sure your device is on the network you meant to be on. Sometimes the fix is sitting right here in the first fact.

Which band, the lane you are in. Your Wi-Fi runs in one of a few separate lanes, and the lane tells you something useful about the connection, though not, by itself, whether the connection is good. Think of them as lanes on a road. The old outer lane is named 2.4. It reaches farther than the others, and it is also the most crowded road in the neighborhood, because every old gadget in every house nearby is crammed into it alongside yours. The middle lane is named 5. It does not reach quite as far, but there is far more room in it, and that room is what you actually feel as speed. The newest lane is named 6. It is the roomiest of the three and it has space for the widest lanes, so if your device and your box both have it, there is a lot of room to work with, though it travels the shortest distance through your walls. And 2.4 is not a failure. It is the right lane for a doorbell or a thermostat, which only ever needed a trickle. It is only a warning sign when the thing sitting in it is the laptop you are trying to work on.

2.4 GHz

- More reach
- Least room
- Often crowded

5 GHz

- Good balance
- More room
- Where most modern devices work

6 GHz

- Most clean room
- Widest lanes
- Shortest reach

Your three lanes, and what each one trades away. 2.4 reaches farthest and has the least room. 5 has much more room and is where most modern devices do their work. 6 has the most room of all and the shortest reach. None of the three is simply the fast one.

The channel, the spot you are sitting in. Inside each lane there are numbered spots, and your box picked one. The screen shows you which. Be careful what you read into that number on its own: **it tells you where you are sitting, not how many neighbors are sitting there with you.** Your phone is not counting the other households on that spot, and on an iPhone it is not allowed to.

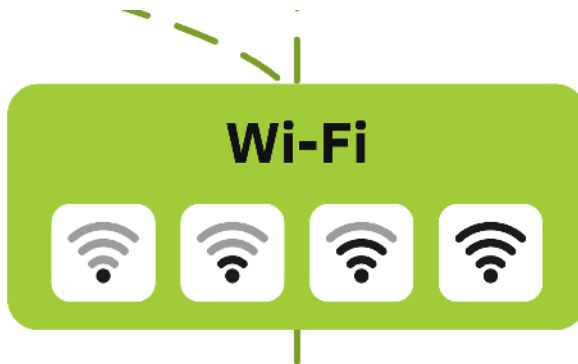
So why does it matter? Because crowding is real, and it is the thing you work out when the other facts refuse to add up. If everyone in the neighborhood has piled into the same spot you are on, everyone takes turns, and taking turns feels exactly like a slow connection, **even when your signal grades Excellent.** That is the combination worth remembering, and it is the one that surprises everyone: **Excellent signal, and still slow.** You will not read crowding off a single number. You will notice it when a connection that looks healthy in every other way is not behaving, and that disagreement is the clue.



Excellent signal + busy channel = still slow

A strong signal is not the same thing as fast Wi-Fi. The signal grade only says how well you hear the box. It says nothing about how many other households are sitting in the spot you are sitting in, and a busy spot slows an Excellent signal down to a crawl. This is the one combination that surprises everyone.

The signal grade, how well you hear the box. This is the one everyone already thinks they understand, the bars in the corner grown up into a real reading. It is also the one place on the whole map where your device puts a number on the air, so it is worth zooming in on. Your device grades how strongly it hears the box on the wall, from **Excellent** at the top, through **Good** and **Fair**, down to **Poor** when you have drifted too far. Excellent means you and the box are having an easy conversation. Fair means you are straining to hear each other across too much distance or too many walls. This matters, a lot, but hold one thought until the Myth-Bust at the end of this chapter: it is one of five facts, not the whole story.



The signal meter, zoomed out of the big picture. This is the Wi-Fi side of that first hop and its four states, from a single dot up to a full meter. It is the same drawing you met in Chapter 1, with the camera pushed in on the one leg that carries a gauge. Nothing further along the road has one.

The link speed, how fast this conversation is actually going. This is the speed your device and your box agreed to talk at, right now, on this connection. It is not your internet plan and it is not a promise on a box in a store. **It is also not the amount of data you are actually moving.** Radio never delivers everything it advertises, and roughly half of this number is what survives as real throughput once the two of them finish taking turns, acknowledging each other and working around the neighbors. Think of it as the rate they agreed to *try* at. A healthy one is a good sign about this connection, and a low one is a strong hint that the radio hop is where your trouble lives.

The five facts

1 Network

am I on the right Wi-Fi?



2 Band

which part of the spectrum?



3 Channel

which spot am I in?



4 Signal

how strongly do I hear it?



5 Link rate

what rate did we agree to try?

The five facts, on one card. Which network you are on, which lane you are in, which spot you are sitting in, how well you hear the box, and what rate the two of them agreed to try. Read these five and you have read your Wi-Fi. There is nothing else to memorize. ###
The other readings on that screen, and why you can walk past them

You will see more than five things on the Wi-Fi Information screen. There is an **RSSI** with a negative number beside it, a **Noise** figure, an **SNR**, and a **Tx Rate**. Nothing in this book asks you to learn them, and here is the honest reason why: they are the raw material the app already used to work out the five plain words you just read. RSSI is the strength reading behind your signal grade. Noise is the background hiss it is competing with, and SNR is the gap between the two, how far your box's voice rises above the room. Tx Rate is the link speed with its engineering name on.

They are on the screen because this app is also used by professionals, and because hiding a number you measured is its own kind of lie. **Read the plain words. The numbers are there when you want them, and they are there for the person you might hand the screen to.**

A good connection, and a bad one

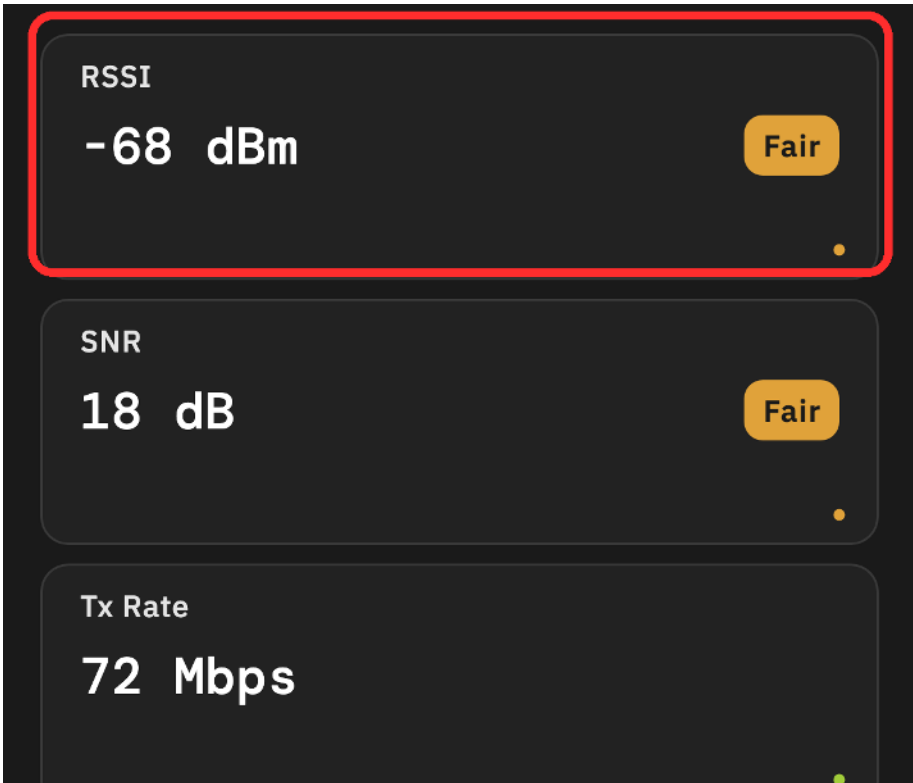
Now look at the two screens side by side, because the shapes are what stick.

The good one reads like a clean story. You are on the network you meant to be on. You are in a fast lane, the 5 or the 6. The spot is not crowded. The signal is Excellent. And the link speed is healthy, because all four of the other facts lined up. Every fact agrees, and the connection feels quick. That is the shape you are looking for, and once you have seen it, you will recognize it anywhere.

The bad one tells on itself the moment you read it. Maybe you are stuck in the slow 2.4 lane. Maybe the spot is crowded, every neighbor stacked on top of you. Maybe the signal is only Fair because you wandered too far from the box. And maybe the link speed down at the bottom is low as well.

Resist the urge to read that as a chain where each one caused the next. It is not. These are five separate clues about one connection, and they do not always move together. A connection can show Excellent signal and a healthy link speed and still crawl, because the spot it is sitting in is packed and everyone is taking turns. **Sometimes all five agree, and**

sometimes they do not, and the disagreement is the clue. One warning sign may be enough to explain what you are feeling. Several may be piling up at once. Either way you no longer need a single magic number, because you have five plain words telling you where to look first.



A bad connection, telling on itself. Stuck in the slow 2.4 lane, sitting in a crowded spot, only a Fair signal, and a slow link speed at the bottom. You can read what is wrong in plain words, with no number to interpret.

This is the gift of the five facts. A slow corner used to be a single useless feeling, the vague sense that the Wi-Fi was bad. Now it is five readable facts, and one or two of them will be the obvious culprit. You stop guessing and start reading. And almost every one of these has a fix you can do yourself, which is exactly where the next chapters go.

What you can do with this already

Walk the five facts the next time something feels slow, in order, and most of the time the problem names itself.

Start with the network. Are you even on the right one? Then the lane. Are you stuck in the slow 2.4 lane when a fast one was available? Then the spot. Is your channel crowded? Then the signal. Is it only Fair because you drifted too far from the box? By the time you reach the link speed at the bottom, you usually already know why it is low, because one of the four facts above it told you. Five plain words, read top to bottom, and the mystery is gone.

Notice what just happened, because it is the same trick as the last two chapters. You did not learn one fact about radios or frequencies or how any of this works under the hood. You learned to read five plain words your device was already showing you, and to know which one is the troublemaker. That is all reading your own connection ever was.

What this chapter cannot do yet is fix any of it. Knowing you are stuck in the slow lane, or sitting in a crowded spot, or straining against a Fair signal is the diagnosis, not the cure. The good news is that the cures are simple, you can do most of them from your couch, and you can watch these exact five facts improve as you do. That is where we go next. You can read your connection now. Next we start moving the readings in the right direction.

Do this now

Two devices, one spot, five facts each. The point is that these facts belong to a conversation, not to your house.

Chapters 1 and 2 already had you walk your home, so leave your feet where they are. This time you are holding the room still and changing the device instead, because that is the one thing the walk can never show you.

1. **Pick one spot and read the five facts on your phone.** Which network, which lane, is the spot crowded, the signal grade, the link speed. Write the five down, or just say them out loud.
2. **Without moving, read the five facts on a second device.** A laptop, a tablet, another phone, anything in the house that can show you the same screen. Same room, same box, same moment.
3. **Compare the two lists.** Most of the time at least one fact will disagree, and often several will. A different lane. A different link speed, sometimes by a lot. Occasionally a different network entirely, which is the surprise worth finding.

Whatever disagrees is the lesson. Same chair, same room, same Wi-Fi, two different answers, which tells you the device is part of the connection too, and not just a passenger. **The five facts describe one device's conversation with the box, not the quality of the Wi-Fi in your home.** So before you change anything about the whole network, find out whether every device has the problem or only one. That single question saves more wasted evenings than any other in this book.

Myth-Bust: “More bars always means better Wi-Fi.”

No. The bars are only one of the five facts, the signal grade, and an Excellent signal carrying nothing is still nothing. You can sit under a perfect, full-bar signal and still crawl, because you are stuck in the slow 2.4 lane, or jammed into a crowded spot with half the neighborhood, or on the wrong network entirely. The bars never knew about any of that. They only ever measured how well you hear the box, not whether the lane was fast, not whether the spot was crowded, not whether you were even on the right network. An Excellent signal is a good thing to have, but it is one fact out of five, and the other four can sink you while the bars sit there looking perfect. So when the bars are full and the Wi-Fi still drags, do not argue with your own eyes. Read the other four facts. One of them is the real story.

Next: Chapter 4 takes the word “slow” apart, because slow how matters. The slow that freezes a video call is a different problem from the slow that drags a big download, and each one has its own fix.

Chapter 4: Slow how? (the four flavors of slow)



“Slow” is four very different experiences wearing one word, and the four send you to four different places to look. That is why “my Wi-Fi is slow” almost never gets solved by the thing people try first. The page that will not load is a different problem from the call that freezes and unfreezes, which is a different problem from the big download that crawls, which is a different problem from the one app that stutters while everything else is fine. By the end of this chapter you will run one screen that grades all four at once, separately, and you will know which one actually bit you. That is the whole game: stop saying “slow” and start saying which kind.

I have spent over twenty-five years listening to people describe slow, and I have learned that the word tells me almost nothing. A friend says the Wi-Fi is slow, and I have to ask four follow-up questions before I know what they even mean. Did a page hang forever? Did your voice on the call chop into robot noises? Did a movie take an hour to download? Or was it all fine except for that one app that keeps spinning? Those are four separate complaints, and only one of them is even about the thing people usually go

reboot. The trouble is that one tired word, “slow,” gets stretched over all four, so everyone chases the same useless fix for four different problems.

Here is the idea most people never get told. There is no single number that tells you whether your connection is good. You have been handed one your whole life, the big download number from a speed test, and it has been quietly lying to you, because it only measures one of the four flavors and stays silent about the other three.

Go back to the map once more, because it explains why. The last chapter zoomed in on the first hop, the one leg of the journey your device puts a meter on. Now look at everything past it. The feed out to your provider, whichever of the ways it travels, and the whole long haul across the world to wherever your thing actually lives. **Not one of those legs has a meter on it.** There is no gauge anywhere out there for you to read, and there never was. So every judgment you make about the rest of the road is made from your end of it, by sending something out and timing what comes back. That is what the six rows in this chapter are: six different ways of doing exactly that, because one way was never enough. So let’s go look at the one screen that measures all four and refuses to blur them into a single happy score.

Open the WLAN Pros Toolbox on your device. Tap **Network Quality**. Don’t read anything into it yet. Just look at how many rows there are.

Transport ● Live · sampling latency ever...



Latency

24 ms

Good



Jitter

6 ms

Good



Loss

0%

Good



Download

180.0 Mb...

Good

Run a test to start tracking

Upload

12.0 Mbps

Good

Run a test to start tracking

Responsiveness

1200 RPM

Good

Run a test to start tracking

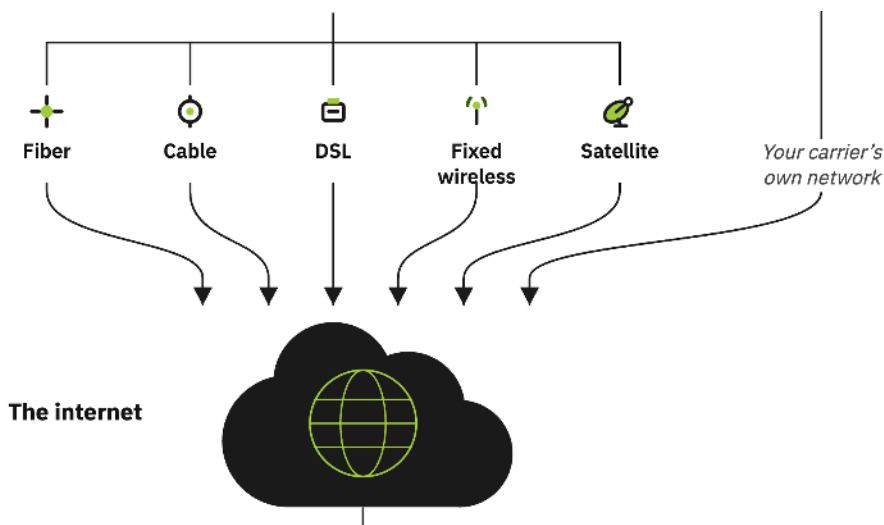
The Network Quality screen, showing six separate readings instead of one big number. Each row is graded on its own. That separation is the whole point.

Six rows, no magic number

The first thing you will notice is what is missing. There is no giant number at the top. No single score, no one percentage, no “your connection is 87 out of 100.” Instead there are six rows, each measuring a different thing, each graded on its own as **Excellent**, **Good**, **Fair**, or **Poor**.

People are so used to the one-number speed test that the six rows feel like the app forgot to add them up for you. It did not forget. It refused. There is a thing I have said in front of a lot of rooms, and it is the most useful sentence in this chapter: the number does not matter, what matters is whether your stuff works. A single score hides exactly the thing you need to see, which is *which kind* of slow you have. Add the six rows into one number and you have thrown away the answer.

Think of it the way a doctor thinks about you. When you walk into the office, nobody hands you one number for “health.” They check your blood pressure, your temperature, your heart rate, each one its own reading, because a fever and high blood pressure are different problems with different fixes. Rolling them into a single “health score” would be useless, and a little dangerous. Network Quality treats your connection the same honest way. Six readings, each meaning its own thing, none of them pretending to be the others.



Past the first hop, no meters at all. Six different ways out to the world, counting the one your phone takes when it leaves your house entirely, and not one of them carries a gauge you can read. This is the same drawing from Chapter 1, with the camera pushed in on everything downstream of the leg that does have a meter. It is the reason a single number was never going to be enough, and the reason there are six rows on the next screen instead of one.

One of the six deserves a sentence of its own, because its name gives nothing away. **Responsiveness** is how quickly the connection reacts when it is already busy. A connection can have plenty of speed and still feel sluggish the moment the house gets going, and that gap is exactly what this row is for. You will see it reported as a number that means nothing to a normal person; ignore the number and read the word beside it, the same way you read the others.

Now here is the part that makes the six rows useful instead of just thorough. Each kind of complaint gives you a row or two worth checking first. You do not have to learn all six names. You have to learn the four feelings, and where each one sends you first. **That is the rule for this whole chapter: the feeling tells you where to look. It does not tell you what you will find.**

IF IT FEELS LIKE THIS → LOOK HERE FIRST

Pages stall, retries help → **Loss**

Calls freeze or go robotic → **Latency · Jitter · Loss**

Big transfers crawl → **Download · Upload**

One app fails, others fine → **Check the app before the network**

The four flavors of slow. The word “slow” splits into four real problems, each with its own feeling, its own row on the screen, and its own fix. Find your feeling in this picture and you already know where to look.

The four flavors, one at a time

Flavor one: the page will not load at all. You tap a link and it just hangs. Not slow, stuck. It spins, maybe it loads halfway, maybe it times out and you tap it again and the second time it works. When a page stalls, half-loads, or only works on the second try, **loss is the first row to check**. It is not the only thing that can do this. A slow name lookup or a struggling server will produce the same feeling, but loss is where to start. Loss means some of the little packages your connection sends out never arrive, so they have to be sent again, and that retry is the pause you feel. On the screen, look at the **loss** row. When loss grades Fair or Poor, this is your flavor. Pages hang, things half-load, and clicking again sometimes fixes it because the second try gets through.

Flavor two: the call freezes and unfreezes. You are on a video call and the other person turns into a frozen statue, then suddenly catches up in a burst of robot voice, then freezes again. Audio chops. Faces stutter and jump. This is the flavor that wrecks meetings, and it has nothing to do with how fast your download is. Check three rows here, not one: **latency**, how long each little package takes to make the round trip; **jitter**, how uneven that timing is; and **loss**, because a call has no time to ask for a missing piece twice. Once a call has enough room to work in, more Mbps does not rescue bad timing or missing pieces. It wants a steady, even beat. When latency is

high or jitter is bad, the beat falls apart, and a call cannot smooth that over. This is the single most misdiagnosed flavor in the world, because the download number can look gorgeous while the call is a disaster.

Flavor three: the big download crawls. You are pulling down a movie, a game update, a giant file, and the little bar inches across like it is wading through mud. This is the job that big Mbps number is best at. If your complaint is that a giant download takes forever, Mbps is exactly the number you care about. It lives in the **download** row, with **upload** as its mirror for sending big things out, like backing up photos or posting a long video. When download grades Poor, big transfers crawl. That is real, and it matters when you move large files. But notice how narrow it is. It is one of four flavors, and it is the only one that big headline number has ever described. Some newer tests do measure latency and loss as well, and ours does. But the number people quote at each other is still the download number, and every time someone reads it out and declares the Wi-Fi good or bad, they are judging all four flavors by the one that happens to be printed largest.

Flavor four: everything is fine except that one app. Every page loads, every call is smooth, every download flies, and yet that one app, that one game, that one website keeps spinning and stalling. Here is the honest answer, and it is the one people least want to hear: this flavor is mostly not on your screen at all. The six rows grade the road, and this trouble is usually not the road. It is somewhere out at the far end you do not own and cannot reach from your couch, or in the app itself on your own device.

So prove it before you touch anything. Try the same service another way, on another device, or in a browser instead of the app. If everything else on the connection stays healthy, start with the app or the service, not with your Wi-Fi. **When five things work and one does not, start by suspecting the one.** The single most expensive mistake in this whole book is rebuilding a network to fix an app that was never using it properly.

Four flavors. Four feelings. Match your feeling to the flavor, and you have already done the hard part, which is naming the real problem instead of the word “slow.”

The call-killer the speed test never sees

Let me show you the trap in one screen, because it is the most important picture in this chapter.

Picture a connection with a beautiful download grade, **Excellent**, the kind of number you would screenshot and brag about. And right next to it, jitter graded **Poor** and loss graded **Fair**. By the only meter most people own, this connection is fantastic. By the meter that matters for a video call, it is a wreck. The download is gorgeous and the call still freezes, because a call never cared about the download. It cared about the steady beat, and the steady beat is broken.

Transport ● Live · sampling latency ever...



Latency

60 ms

Fair



Jitter

45 ms

Poor



Loss

2%

Fair



Download

480.0 M...

Excellent

Run a test to start tracking

Upload

240.0 M...

Excellent

Run a test to start tracking

Responsiveness

1200 RPM

Fair

Run a test to start tracking

The call-killer profile. A gorgeous Excellent download grade sitting right next to Poor jitter and shaky Fair loss. A speed test would call this connection excellent and send you off happy, while your next video call freezes solid. The download was never the part that mattered for the call.

This is the exact connection that fools everyone. The person runs a speed test during a frozen call, sees a big healthy number, and concludes the Wi-Fi is fine and the call software is broken. Half right. The connection had a great download and a broken beat, and the one meter they owned could only see the half that looked good. **That does not clear the call software, and it does not convict it either.** What it does is hand you a reason the call may be struggling, which is a great deal more than you had. Network Quality shows you both halves, side by side, so you can point straight at the row worth chasing. For a frozen call, look first at latency, jitter and loss, not at the download number.

Once you have seen this picture, you cannot unsee it, and you will never again trust a single download number to tell you whether your connection is good.

Do this now

Run Network Quality twice, and watch the rows move on their own, the same way the two roads moved on their own back in Chapter 1.

1. **Run it once, sitting still, with nothing else going on.** Read all six rows. Do not add them up. Just notice which grade Excellent or Good and which do not. This is your baseline.
2. **Now make the connection work hard, and run it again.** Start a large download or upload, kick off a cloud backup, or get several devices busy at once. A single movie stream may not be enough to make a modern connection breathe hard. Then run Network Quality while that is happening. **Watch which rows change, without deciding in advance which ones should.** You may find the big Mbps number still looks healthy while latency, jitter, loss or responsiveness gets worse. That is the whole point: a connection has to work when you are using it, not only when the house is quiet.
3. **Name your real complaint.** Think of the last time your connection felt “slow.” Which of the four flavors was it actually? Page hanging,

call freezing, download crawling, or one app stalling? Whatever it was, you now know which row to look at, and “slow” has stopped being one word and become four real problems.

Two runs, and you will never again accept “slow” as an answer, from yourself or anyone else.

Myth-Bust: “A big speed-test number means my connection is good.”

No. That number proves you moved a lot of data during the test, which is genuinely useful and is exactly the right meter for one situation: pulling down a movie or a giant file. What it does not prove, on its own, is that pages will not hang, that calls will not break up, or that anything will feel quick while the house is busy. Some newer tests do report timing and loss alongside the big figure, and ours does, but the number people quote at each other, and the one printed largest, is still the download. You can run a test, see a giant glorious number, feel great about it, and then watch your very next video call freeze into a statue. The big number was real. It was just answering a question you were not asking. The number does not matter on its own. What matters is whether your stuff works, and “your stuff” is four different things. So the next time someone runs a speed test and declares the Wi-Fi good or bad, you can smile, because you know they just judged a four-part exam by reading one answer.

Next: Chapter 5 takes everything you have learned to read and turns it into action. You will run the app before a fix and after a fix, and watch the grade actually move, so you finally get the one thing home Wi-Fi has never given you: proof that what you did worked.

Chapter 5: The fixes you can actually do yourself



Before you buy anything or call anyone, there are a few plain things worth trying, and you can do all of them from your own home in the next ten minutes with no tools and no technician. **They are in order on purpose, gentlest first.** Reconnect the one device. Move closer and see what changes. Check that you are on the network you meant to be on. Get the box up off the floor and out of the cabinet. And if it comes to it, reboot the right box, once. And the reason these fixes have sat there unused for years is not that they are hard. It is that nobody ever gave you a way to see whether they worked. This chapter hands you that missing piece. You will read a grade, do one fix, read the grade again, and watch it move. The app becomes the thing home Wi-Fi has never had: a feedback loop.

Here is something I have come to believe after a long time fixing this stuff. The reason so much home Wi-Fi is bad is not that people are careless or dumb. It is that they have never been punished for the small mistakes, and never rewarded for the small fixes. You buried the box in a cabinet and the Wi-Fi still sort of worked, so nothing told you that you had just thrown away half your signal. You stood three walls away and it still sort of worked, so

nothing told you that one step closer would have fixed it. Wi-Fi is forgiving in the worst way. It keeps limping along when you do it wrong, so you never learn. The app on your device changes that, gently. It shows you the grade before, you make a change, and it shows you the grade after. For the first time, the Wi-Fi tells you the truth about what you just did.

Before you touch anything: do not make it worse

Everything in this chapter is small, reversible, and safe to undo. That is deliberate, and it is worth saying plainly what is *not* on the list, because a frightened person following advice from a stranger on the internet does these things and turns a ten-minute annoyance into a ruined evening.

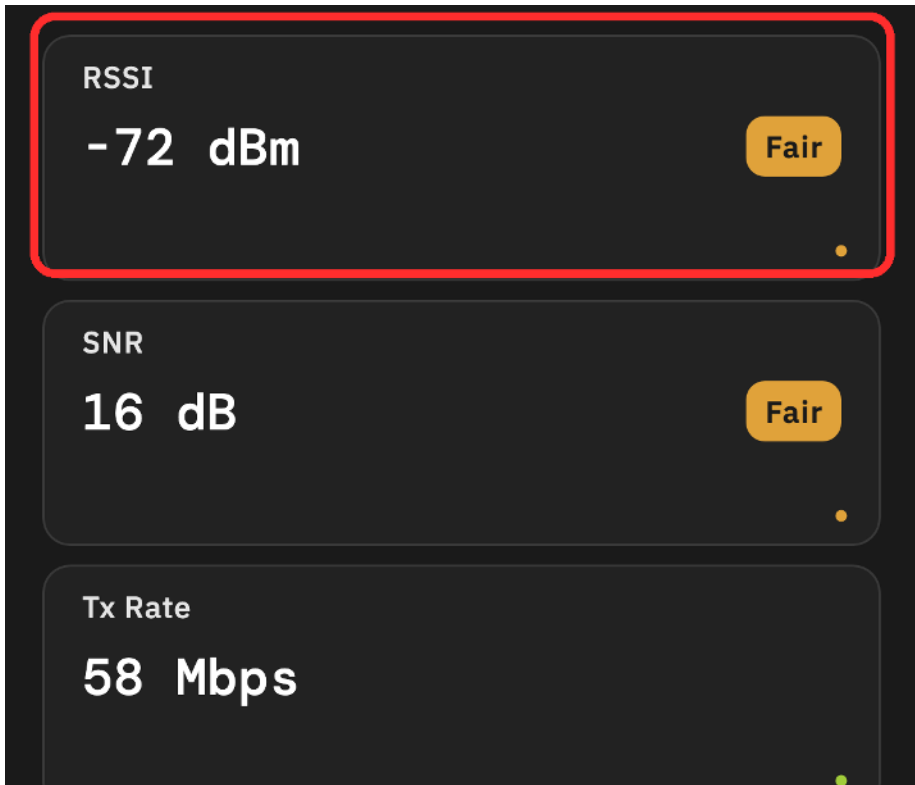
Do not press the little pinhole reset. Do not factory-reset the box. It wipes your network name, your password, and every setting your provider put there, and now nothing in the house can connect, including the device you were using to look up the advice.

Do not start changing channels or security settings because a forum post told you to. Those settings are not why your back bedroom is weak.

Do not unplug the thin glass fiber line if you have one. It is not a cable you reseal, and the connector is easy to ruin.

One reversible change at a time, then read the grade again. That is the whole method, and it is also the protection: if a change did nothing, you put it back and you have lost nothing but a minute.

So before any fix, you take a reading. After the fix, you take it again. Two screens hold this whole chapter together. **Wi-Fi Information** shows you the live grade of the radio hop, the one you read about back when you learned the five facts. **Test My Connection** gives you the plain one-tap verdict on whether the whole thing got better. Run one or both before, do the fix, run them again after. Watch the grade move. That is the loop. Everything below is just five good reasons to run it.



The “before” reading. Fair signal, the slower network, a low link speed. This is your starting line. Do not skip it. If you do not know where you started, you cannot prove you improved.

Fix one: reconnect the device

Start here, every time, because it is the smallest thing you can do and the easiest to undo. **Turn Wi-Fi off on the one device that is misbehaving, wait a few seconds, and turn it back on.** Not airplane mode, because that moves more than one thing at once, and this whole chapter is about moving one thing at a time.

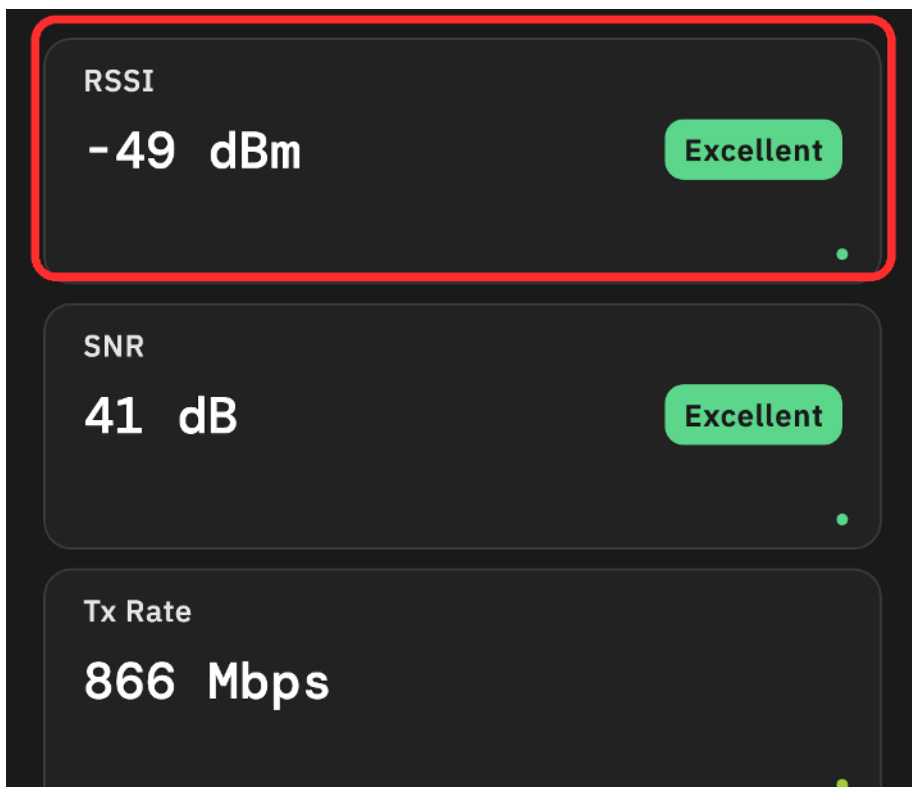
Phones and laptops are stubborn. Once they latch onto a box they hang on long after you have walked into another room with a closer, stronger box sitting right there. Turning Wi-Fi off and on breaks that grip and lets the device choose again from where you are standing now, rather than from wherever you were when it first connected.

No furniture moves. Nothing gets unplugged. Nobody else in the house loses their connection. **Read the grade, flip it off and on, read the grade again.** If it climbed, you are done in fifteen seconds, and you never had to touch the box on the wall at all.

Fix two: move closer, and find out whether distance is the problem

The single highest-yield thing you can do for bad Wi-Fi is also the dumbest-sounding. Get closer to the box. Walls, floors, furniture, and distance all chew up the radio signal between you and the box on the wall, and the further away you sit, the more gets chewed up. Half the “slow Wi-Fi” in the world is just somebody sitting too far from the box with too many walls in between.

You do not have to believe me. Take the before reading on Wi-Fi Information right where you normally sit. Note the signal grade. Now walk into the same room as the box, or just halfway there, and read it again. Watch the signal grade climb. That is distance, made visible. **And notice what this really tells you: if the reading improves a lot when you move, distance or what is sitting between you and the box is part of your problem.** That is a diagnosis, and it is worth more than the walk itself, because sitting next to the box forever is not a fix for a home office in the back bedroom.



The “after” reading, from closer to the box. Same screen, same network, a better grade and a faster link. Put it next to your “before” reading from earlier and you are looking at proof the fix worked, in your own home, in your own words.

THREE WALLS AWAY

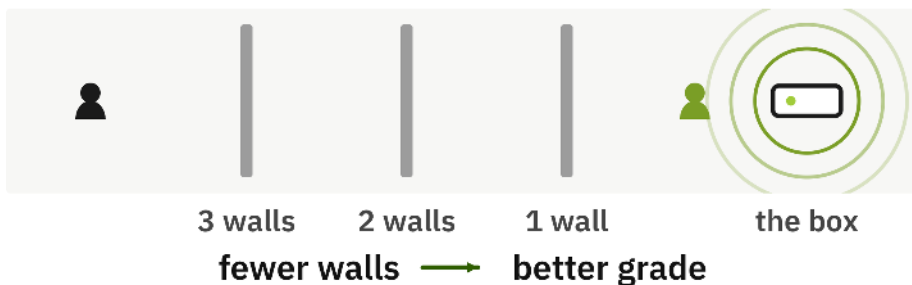
Weak

Straining to hear across the house.

SAME ROOM

Strong

Few walls, an easy conversation.



Get closer. Same box, same phone, same person. The only thing that changed is how many walls and how much air sit between you and the box. The grade follows the walls. Fewer walls, better grade.

And it is not only how far away you are. What sits between you and the box matters just as much. A thick wall, a brick chimney, a mirrored door, a metal filing cabinet, a refrigerator, a fish tank full of water. Any of those can take a real bite out of the signal, and how big a bite depends on what it is actually made of and how squarely it sits in the way. You can be close to the box and still get a bad grade because a refrigerator is standing in the way. That test takes ten seconds. Read the grade where you are, take one step sideways so the big metal thing is no longer between you and the box, and read it again. If the grade jumps from one small step, you just found a blocker.

Water and metal are usually the ones that surprise people, which is one reason kitchens and bathrooms turn up as dead spots so often, though your house is its own experiment and the only way to know is to read it. You cannot move the chimney. But you can move your chair a foot, and you can stop being baffled that one specific spot in your home is always terrible.

Now, you cannot rearrange your whole life around the box either. But you can learn where your home's dead corner is, and you can stop being surprised when the far bedroom is the worst room in the house. And if the room where you actually work or watch movies is the dead corner, that is your clue that the box itself is in the wrong place, which is the next fix.

Fix three: make sure you are on the network you meant to be on

Most homes are quietly running two or three networks at once, and a lot of devices sit on the slow one out of pure habit. Back when you read the five facts, you met the bands, the lanes on the road: the older 2.4 GHz lane that is crowded and slow but reaches far, the 5 GHz lane that is much faster, and on the newest boxes the 6 GHz lane that is faster still over shorter distances. A device stuck in the slow lane feels slow even when everything else is perfect.

Run Wi-Fi Information and read which network you are on. **First question: is it the one you meant?** The guest network, an old name you

stopped using, a neighbor's network you joined once, a phone hotspot still in range. That alone is the fix more often than people expect.

Second question, and only if your home deliberately has more than one name. If it shows the 2.4 GHz lane and you are anywhere near the box, and there is a separate faster name to join, joining it is worth a try. **If your box uses one name for all the bands, which most modern ones do, leave it alone and let your device choose.** Sometimes the faster lane is a separate network name with “5G” or “5 GHz” in it, and you simply pick that one in your device's Wi-Fi list and forget the slow one. Sometimes the box hands both lanes out under one name and quietly parks you on the slow one. Either way, read the before, switch to the faster network, read the after, and watch the link speed jump.

There is one honest catch, and it is the reason the slow lane still exists. The faster lanes do not reach as far through walls. So the right move is the faster network when you are close enough, and you will know you went too far when the grade starts to fall. Get on the fast lane, stay on it until the walls make you drop back. The app tells you exactly where that line is.

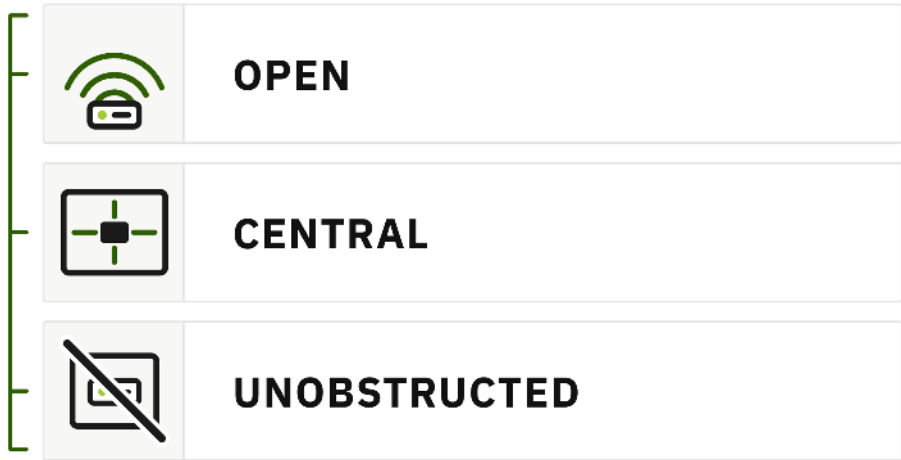
Fix four: get the box up and out

Where the box lives matters as much as where you sit. And almost everyone hides their box in exactly the wrong spot. On the floor. In a cabinet. Behind the television. Down in the basement next to the cable that comes in. People tuck it away because it is ugly, and every place they tuck it strangles the signal.

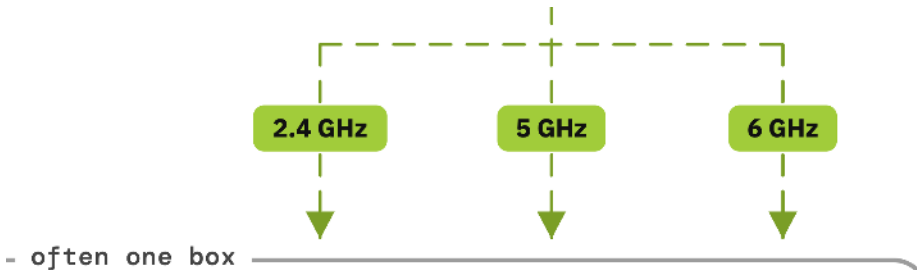
What the box puts out goes mostly sideways, out into the rooms around it. Put it on the floor and you have aimed half of that into the concrete. Bury it in a wooden cabinet and you have wrapped it in a muffling blanket. Shove it behind a big metal-backed television and you have parked a wall right in front of it. Each of these quietly costs you signal you paid for and never got.

The fix is plain, and it is three words: **open, central, unobstructed.** Get the box up off the floor and out into the open air where it can breathe. Out of the cabinet. Away from big metal. Not behind the couch. Height can help, but it is the least important of the three. An open shelf in the middle of the house beats a high corner behind a television every time. You do not need to mount it perfectly. You just need to stop actively smothering it. Take the

before reading, move the box up and out, then take the after reading and watch the grade improve across the rooms you care about.



Where to put the box. Three requirements, not a ranking: open, central, unobstructed. Almost everyone hides it low, boxed in, behind metal. You are not decorating. You are getting out of your own signal's way.



The three lanes, zoomed out of the big picture. Same drawing as Chapter 1, camera pushed in on the part where your box hands out three different roads through the air. Two of the five fixes in this chapter are nothing more than getting yourself onto a better one of these.

Fix five: reboot the right box, once

Now the famous one. Rebooting. It does sometimes help, which is exactly why it has become the thing everyone tries first, last, and always. But here is what has gone wrong. People reboot the box that was working fine, they reboot it over and over, and they reboot it before they have any idea whether the box was even the problem. That is a nervous habit.

Do it right and do it once. First, read the grade and find out whether this is even a Wi-Fi problem. If Test My Connection pointed at one specific app, rebooting your Wi-Fi equipment is unlikely to help. If it pointed at the internet feed, do not assume the Wi-Fi radio is the thing that needs restarting. Work outward instead, and you can stop now and save yourself ten minutes. If it really is the connection, then reboot, and reboot the right thing. In many homes there are two boxes: the one your provider's cable plugs into, and the box that sprays out the Wi-Fi, and sometimes they are the same single box. **Unplug the power cable, not the internet cable, and never the thin glass fiber line if you have one.** Wait until it has fully gone dark and quiet, then plug the power back in and let it fully wake up before you judge anything. **And never use the little recessed RESET button to do this.** That is not a reboot, it erases everything, and you will spend your evening putting the house back together. One clean reboot. Then take the after reading and see whether the grade actually moved. If it did, good. If it did not, rebooting was never the answer, and now you know, instead of doing it a fourth time and hoping.

- 1 Reconnect the device**
Wi-Fi off, wait, on. Not airplane mode.
- 2 Move closer**
and find out whether distance is the problem
- 3 Check the network**
are you on the one you meant?
- 4 Get the box up and out**
open, central, unobstructed
- 5 Reboot the right box, once**
power cable, never the reset pinhole

The fix ladder, gentlest first. Start at the top and work down, and take a reading between every rung. Most homes never need to climb past the third one. The whole point of the app is that you can see exactly which rung fixed it.

Optional: when a separate network for the small stuff helps

This one is not part of the ladder, and most people should leave it alone. It is here because it genuinely helps in one specific situation, and because you may be told to do it by someone who does not know your house.

If your box already uses one network name across 2.4, 5 and 6 GHz and your devices are working, do not split it because of this book. One name is how modern home Wi-Fi is meant to work. Your phone and laptop move between the bands on their own, and they do it well when left alone. **Splitting the fast lanes apart is a popular piece of advice that makes things worse,** because it takes a decision your device was making correctly and hands it to you.

The one case worth the trouble is the **older gadget that will not join at all**, or keeps dropping: a thermostat, a doorbell, a plug, a light bulb, an older printer. Some of those only ever spoke 2.4 GHz, and a few of them get confused when a network offers more than that. **Giving 2.4 GHz its own name gives those devices somewhere simple to live,** and it is a real fix for a real problem.

What it costs you, so the decision is yours and not a surprise later: another name and password to keep track of, a device that can end up stuck on the slow name after you carry it across the house, and one more thing to explain to anyone else who lives there.

When moving the box is not enough

There is an honest answer this chapter owes you, because the fixes above assume one box can reach everywhere it needs to, and in plenty of homes it simply cannot. A long house, a house on three floors, a detached office over the garage, thick old walls, and sometimes the far room is beyond the reach of one well-placed box no matter how open and central you make it.

If your walk-closer test says the signal is fine near the box and poor where you actually sit, and moving the box helps one room while hurting another, **you have stopped having a settings problem and started having a coverage problem.** The answer to that is another Wi-Fi unit, properly

placed, a mesh node or a second access point. It is not a bigger internet plan, which buys you nothing on this road at all.

One caution, because it is the most common way people make this worse. Do not grab the cheapest range extender and plug it in halfway down the hall. A badly placed extender repeats a weak signal and adds a second weak hop on top of it, and you end up with more bars and less working Wi-Fi. If you are adding equipment, placement is the whole job, and that is the point where this stops being something you do from the couch. Chapter 6 is about knowing when you have reached that line.

Do this now

Pick one fix and prove it works on your own connection. This is the whole loop in five minutes.

- 1. Take the before reading.** Sit where the Wi-Fi annoys you most. Run Wi-Fi Information and note the signal grade and which network you are on. Run Test My Connection and note the verdict. Write both down. This is your starting line.
- 2. Do exactly one fix.** Just one, so you know what caused the change. Get closer, or move the box up and out, or join the faster network, or step out from behind the blocker. One change only.
- 3. Take the after reading, and keep everything else still.** This is the part people get wrong, and it is the difference between an experiment and a guess. **If you moved the box, read from the same chair you read from before.** If the thing you are testing is your own position, then your position is the one thing allowed to change. Change the box and the chair at the same time and you have learned nothing about either. Run the same two screens and compare. Did the signal grade climb? Did the verdict improve? Watch the grade move. That movement is the proof, and it is the thing you have never had before.
- 4. Climb the ladder only as far as you need.** If one fix solved it, stop. You are done. If it did not, do the next fix and read again. Most homes never get past the third rung.

One fix, one before, one after, and you have done the thing nobody ever taught you to do: you changed something and you measured whether it

helped.

Myth-Bust: “Rebooting always fixes it.”

No. Rebooting fixes some things some of the time, and the rest of the time it is ten minutes of your life you will never get back. The trouble is that rebooting is the one move everyone knows, so it gets used for every problem, including all the problems it cannot touch. Reboot the box and it does nothing for the wall between you and it. Nothing for the refrigerator in the way. Nothing for the box stranded down in the basement. Nothing for a tired server out in the world having a bad day. People reboot because it is the only move in their pocket, and then they reboot again, and again, because they cannot see whether it helped, so they keep trying the one thing they know. That is the real damage of “reboot always fixes it.” It crowds out the four fixes that would have actually worked, and it sends you in circles because you have no way to tell whether anything changed. So do it right. Read first. Find out if it is even a Wi-Fi problem. If it is, reboot the right box, once, and then read the grade to see if it moved. A reboot you can measure is a fix. A reboot you just keep doing is a worry bead.

Next: Chapter 6 covers the problems that are not your Wi-Fi at all, the ones where the honest move is to stop fixing and pick up the phone, and exactly how to tell when you have crossed that line.

Note: Flip it off and back on, and let it re-choose.

The gentlest fix is Fix one, and it is worth repeating here because people skip it. Turn Wi-Fi off on the one device, wait a few seconds, turn it back on, and read the grade again. Fifteen seconds, nothing unplugged, nobody else in the house disturbed.

Chapter 6: When it's not the Wi-Fi at all



Sometimes the most useful thing your device can tell you is to stop changing the Wi-Fi. That sounds like bad news. It is the opposite. The fixes in the last chapter are real, and they solve most of what goes wrong at home. But a fair number of slow days are not your Wi-Fi, not your job, and not yours to fix from the couch. The skill this chapter teaches is recognizing those days the moment they arrive, so you stop pouring effort into a problem that was never yours. By the end you will know the four times to put the phone down, and exactly who to hand the problem to.

I have watched people fight problems they could not win for an entire evening. They moved the box. They unplugged it, counted to ten, plugged it back in. They walked around the house holding the phone in the air like they were trying to get a signal on the moon. And the whole time, the thing that was actually slow was sitting in a building three states away, on a server they will never touch, having a bad night all on its own. Every minute they spent was wasted, because they never stopped to ask the one question that ends the fight. Is this even mine to fix?

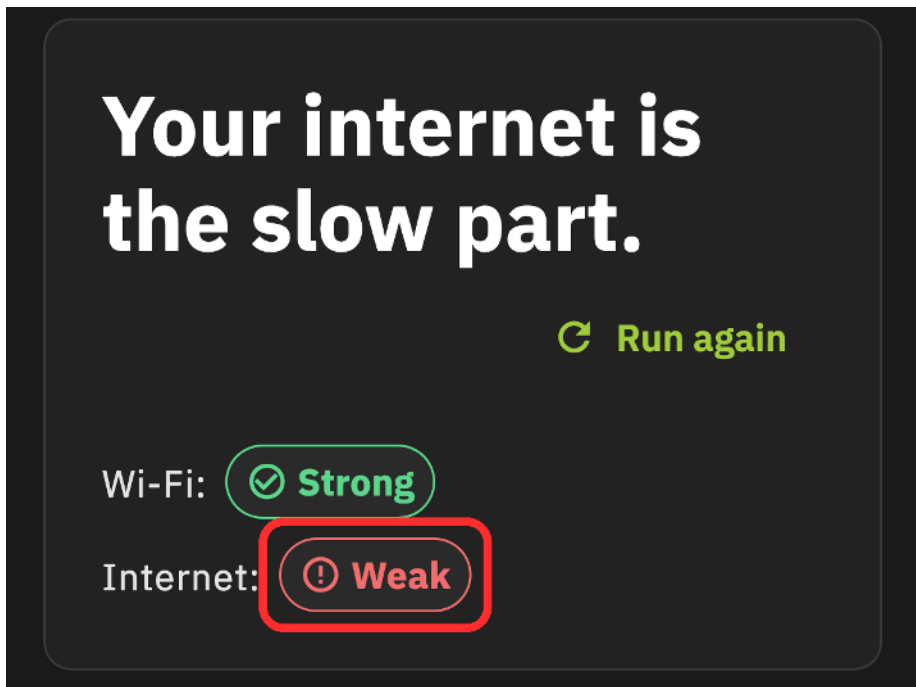
There are four times the answer is no. The internet feed coming into your home is dead or weak. A sign-in page is waiting for you and you never finished it. One app or one website is broken while everything else is fine. Or the network is bigger than a home, and it needs a real professional. Each one has a tell, and your device shows you the tell. Once you can read it, you stop wasting evenings. So let's go learn the four handoffs.

The same tool, read for the door instead of the fix

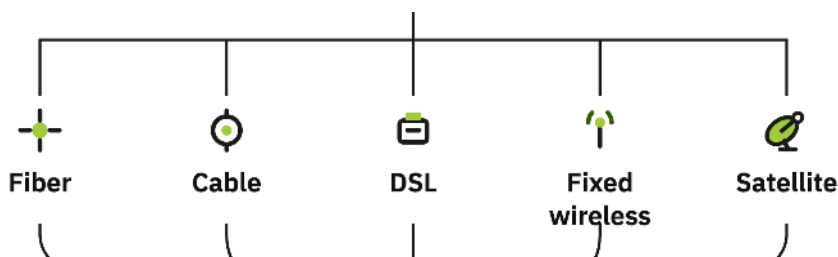
Open the WLAN Pros Toolbox and tap **Test My Connection**, the same screen from Chapter 1. You already know how to run it. This time you are going to read it differently. In Chapter 1 you read the result to find the slow road. Now you are going to read it to find the door, the moment it tells you to stop and hand off.

Run it.

When the two results settle, two of the four outcomes are not instructions to *you*. They are instructions to someone else, and your device is handing you the proof to give them.



Internet the weaker road. In Chapter 1 this told you the slow part. Here it tells you something more useful: stop working. Standing near the box does nothing. This is a call to your provider, and your device just gathered the evidence for the call.



How the feed actually reaches you, zoomed out of the big picture. Fiber, cable, DSL, fixed wireless, satellite. This chapter lives out here, past your box and past anything you own, which is exactly why none of the fixes in the last chapter would have touched it.

Door one: the problem points past your Wi-Fi

When the Internet result is the weaker grade, or worse, when it reads Weak while your Wi-Fi reads Strong, you have hit the most common false alarm there is. The radio hop in your home looks like it is doing its job, and whatever is holding you back is further out. **Where exactly, this reading cannot tell you.** It could be the cable into the building, the equipment it lands on, or something out past that on your provider's side, but it can tell you the direction to work in, and that is worth a great deal.

So stop reaching for the fixes in the last chapter. Every one of them improves the radio hop, and the radio hop is not where the evidence is pointing. Before you pick up the phone, do one more thing: run it again, and if there is a second device in the house, run it there too. **One reading is a hint. The same reading twice, on two devices, is evidence.**

This is the moment people waste the most time. They feel slowness, they blame Wi-Fi out of habit, and they start rearranging furniture. Your device just told you not to. The problem is on the cable coming into the building, or out past it, on equipment your provider owns and you do not. There is one move here, and it is a phone call.

Make the call. And here is the part that changes how the call goes: you are not calling to say “my internet feels slow.” You are calling with a reading. More on exactly what to say in the next chapter, but the short version is that

a person holding a measured result gets taken seriously, and a person describing a feeling gets told to reboot and call back.

Door two: a sign-in page is waiting, and you never finished it

This one is not about a result on the screen. It is about something you saw earlier and walked away from. You joined a Wi-Fi network at a hotel, a coffee shop, an airport, or the guest network at an office, and a web page appeared asking you to type a room number, accept the terms, or click a button that said “connect.” And you got distracted, or you closed it, or you assumed it would sort itself out. It will not. Nothing works until you finish that page.

Here is the part almost nobody understands, and it is the single most freeing idea in this whole book.

If a network ever asked you to sign in on a web page, your Wi-Fi already worked.

Read that again. That sign-in page did not arrive by magic. It traveled to your device over the very Wi-Fi you are about to blame. The radio hop carried it to you. The connection to the box succeeded. The only thing standing between you and the internet is a page you have not finished, a button you have not pressed, a box you have not checked. The radio carried that page to you, so it was working at least well enough to do that much. Finish the sign-in before you troubleshoot anything else. Go back to that page and complete it.

I have said this on stage more times than I can count: if a sign-in page appeared, the Wi-Fi already worked well enough to carry you to that page. Finish the sign-in before you start blaming the Wi-Fi. The page is the step you skipped, and finishing it is usually the whole fix.

Door three: one app, one website, one bad night

When both results read Strong, you already know this answer from Chapter 1, and it bears repeating here because it is a handoff too. Both roads are healthy. The slow thing is one app, one website, or one server out in the world having a rough night. So do not start changing your Wi-Fi to fix one

app. Try another app, or the same service on a different device. If everything else keeps working and only the one thing fails, start with the one thing.

The handoff here is not to a person. It is to time, or to a different app. The fix is to wait it out, or to do the same thing somewhere else. If one streaming service is buffering and everything else plays fine, the problem is that one service, not your home. If one website spins and ten others load instantly, that one website is having the bad night, not you. Switch apps, switch sites, or come back in twenty minutes. Very often it comes right on its own, and you did not have to touch a thing.

Door four: the network was designed, and not by you

The last door is not about the building. It is about whether the network was **designed**.

A five-person office with one box from the provider is a home, for every purpose in this book. So is the dentist, the coffee shop, the little shop on the corner. One box, a few rooms, a handful of people leaning on it. Every chapter you have read applies, and you should go and use it.

The line is crossed when several boxes were deliberately put in place to work together. That is a designed system: an office with access points down a hallway, a school, a clinic, a hotel. Somebody chose where each one went, what channel it sits on, how much power it puts out, and how a laptop should hand off from one to the next while its owner walks down the corridor. **You can still measure it.** Run everything in this book, gather the readings, write them down. **What you must not do is start changing it:** moving a box, renaming a network, unplugging something, pressing a reset. Those changes do not stay where you put them.

This is not a failure on your part. It is the honest edge of the do-it-yourself line. You can still run Test My Connection, and you absolutely should, because the reading you hand to the office IT person or the installer is the same gift it is to the provider. But the fixing is theirs. Moving a box in an office can make things worse for fifty other people, because in a real network the boxes are placed to work together, and yanking one out of position breaks that on purpose.

YOU CAN KEEP WORKING HERE

- 1 Reconnect the device
- 2 Move closer
- 3 Check the network
- 4 Get the box up and out
- 5 Reboot the right box, once

STOP CHANGING THE WI-FI

- internet side stays weak
- one service fails
- sign-in waiting
- designed multi-AP network

Stopping is not giving up. It is part of the diagnosis.

The do-it-yourself line. The left side is yours, and the last chapter taught you all of it. The right side is not, and the skill is knowing the instant you have crossed over. Your device shows you the line. Test My Connection's "Internet weaker" and "both strong" results are both on the right.

When they use a word you don't know

There is one more thing your device gives you for the handoff doors, and it is for the conversation itself. When you call the provider, or talk to the office IT person, or the landlord's installer comes by, they will use words. Band. Channel. Latency. Throughput. The names of things, said quickly, as if you should already know them. You do not have to.

Open the WLAN Pros Toolbox and tap **Glossary**. Type the word they used, say they said "channel," and the screen filters straight to it. You get a plain-English meaning, no decibels, no math, just what the word means in a sentence you can read. Do it right there in the conversation if you need to. There is no shame in looking up a word. Most people using it are not trying to talk over you. It is simply the vocabulary they work in all day. Look it up, understand what they meant, and stay in the conversation.

Plain-language definitions of 92 Wi-Fi terms across 8 categories. Search by term, abbreviation, or any word in a definition.

Definition language

English



🔍 channel

Bands & Spectrum

8

2.4 GHz Band

The oldest Wi-Fi band. It travels farther and through walls better than higher bands, but

The Glossary, with a word a provider or installer just used. Here it is “channel”: type the word, read the plain meaning, stay in the conversation. Your device hands you the vocabulary so nobody can talk over your head.

A handoff is not a failed repair. A good diagnosis sometimes ends with knowing exactly who should work on the next part, and being able to hand them the readings. You did not fail to fix the Wi-Fi. You found out it was not the Wi-Fi, which is the harder thing and the more useful one.

Putting the four doors together

Step back and look at what you can now do. Most of this book taught you to fix things. This chapter taught you the opposite skill, and it is just as valuable: knowing when not to. A dead feed is a phone call. An unfinished sign-in page is a page to go finish. One broken app is a wait or a switch. An

office network is a professional's job. Four doors, each with a tell, each showing on the device in your hand.

The reader who knows these four doors stops losing evenings. They stop rebooting boxes that were fine. They stop moving furniture to fix a server three states away. They make the right phone call the first time, holding the right evidence. That is not a smaller skill than fixing. On a lot of nights, it is the bigger one.

Do this now

The goal here is to practice recognizing a handoff before it happens, so you catch the next one in real time.

1. **Run Test My Connection right now and decide which side of the line you are on.** If both results are Strong, the evidence is not pointing at your connection tonight. Sit with that. That is the answer most people refuse to believe.
2. **Find the last sign-in page you skipped.** Think back to the last hotel, airport, or coffee shop. Did a page ask you to accept terms or type a room number? Picture it. That page was proof your Wi-Fi worked. Carry that idea forward: next time, go finish the page first, because finishing it is the fix.
3. **Look up one word in the Glossary.** Pick any word a tech has used on you that you nodded along to without knowing. Type it in. Read the plain meaning. Now you own that word, and the next time someone says it, you will not flinch.

Three small habits, and you will never again spend an evening fighting a problem that was never yours.

Myth-Bust: "If the website is slow, the Wi-Fi is bad."

No. One slow website is one slow website. When both results read Strong, your Wi-Fi carried your request to the box, your internet carried it out to the world, and both did their jobs. The slow part is sitting on a server you do not own, in a building you will never visit, having a bad night entirely without your help. You can test this in ten seconds: open a couple of unrelated sites or apps. If it loads fast, the first one was the problem, not your connection. People blame the Wi-Fi because it is the part they can see and the part they think they can fix. But the connection already delivered. And here is the friendliest version of the same truth, the one to keep in your pocket forever: if a network ever asked you to sign in on a web page, your Wi-Fi already worked. The page proved it.

Next: Chapter 7 turns these readings into a short, credible script. When you hand off to the provider, the installer, or the office IT person, you will hand over facts, not feelings, and you will be taken seriously the first time.

Chapter 7: Talking to the people who can fix it



When the problem is not yours to fix, the most powerful thing you can do is hand over a fact instead of a feeling. That is the whole chapter, and it is the difference between being taken seriously and being told to reboot and call back. Support hears “it’s so slow” a hundred times a day, and there is almost nothing in that sentence to work with, so the call has to start at step one every time. A person who arrives with three measured readings copied straight off their phone has handed over somewhere to begin. By the end of this chapter you will be that person, and you will close this book holding everything you need to fix what you can and credibly hand off what you can’t.

I want to tell you a short story about an airplane, because it explains why a few plain facts beat all the frustration in the world.

In nineteen thirty-five the Army was testing a new bomber at Wright Field, bigger and more complicated than anything before it. On a test flight it climbed, stalled, and crashed, and two of the five men aboard died. Nothing was wrong with the airplane. Someone had left a control lock engaged. **The problem was never that the airplane needed a better pilot. It**

needed a better procedure. So the Army did not send its pilots back to school for years. It wrote a checklist, a short list of things to verify before takeoff, and those bombers flew more than nine thousand hours after that without a major accident. Decades later surgeons borrowed the idea, and in eight hospitals around the world a nineteen-item checklist cut complications from eleven percent to seven. A checklist takes something complicated and frightening and turns it into a few facts, checked in order, that anyone can hand to anyone.

That is exactly what your device has been building for you, chapter by chapter. A short list of facts. The Wi-Fi grade. The internet grade. The thing that was actually slow. You do not have to understand the radio any more than a pilot has to rebuild the engine. You just have to hand over the facts, in order, to the person who can act on them. So let's learn how to hand them over.

Copy the fact, don't describe it

Here is the move that changes every one of these conversations. On every result screen in the WLAN Pros Toolbox, there is a way to copy what you are looking at. Test My Connection, Network Quality, Wi-Fi Information: each one lets you copy the result you just ran, word for word, exactly as the app stated it.

Run whichever test fits the problem. Then copy the result.

Test My Connection



**Your internet is
the slow part.**

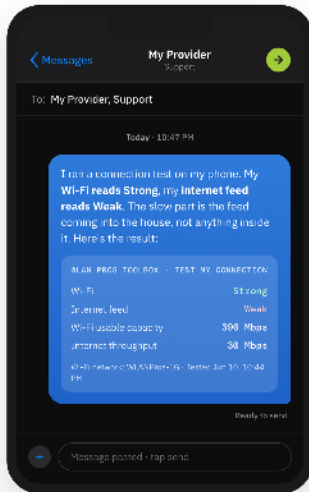
 **Run again**

The copy action, sitting on a result screen. One tap takes the whole reading, exactly as the app wrote it, and puts it on your clipboard. No retyping, no remembering numbers, no getting it slightly wrong. The fact, captured.

Now open your messages, or your email, or whatever you use to reach the provider or the installer, and paste it in. That is it. The reading you ran lives inside your message now, in the app's own words, and you did not have to translate anything or remember a single grade.

Paste the fact, not the feeling

The reading you ran lives inside your message now, in the app's own words. No retyping, no remembering numbers, no getting a single grade wrong.



1

Facts, in your phone's words, not your frustrated ones.

The same reading you ran, pasted straight into a message to your provider. This is what credible looks like: the actual result, in the app's plain words, instead of "it's so slow."

It already did half their job. It ruled out the inside of your home and pointed at their feed. You just handed over a fact.

The whole script State what you measured, paste the proof, stop talking. The reading is enough, because the reading is true and you can prove it.

Figure 516. The copied reading, pasted into a message to your provider, ready to send.

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The same reading, now pasted into a message to your provider. This is what credible looks like: the actual result, in the app's plain words, ready to send, instead of "it's so slow." You just handed over a fact.

Why the fact wins

Picture the two messages side by side. One says "my internet is so slow, please fix it!!" The other says "I ran a connection test twice. My Wi-Fi side reads Strong and the internet side reads Weak. I have pasted the results below." Imagine you work the help desk. Which message do you take seriously? Which one can you actually act on?

The first message is a feeling. There is very little anyone can do with a feeling except start at the beginning, and the beginning is "have you tried rebooting." The second message is something measured, and it has already done a real piece of the work. **It says your Wi-Fi side looked healthy**

while the internet side did not, and it says you saw that more than once. That is a measured starting point rather than a guess, and it is the difference between someone working through a list and someone investigating a lead. **You did the triage. You did not diagnose their network, and you do not need to.**

FROM FEELING TO EVIDENCE

Less useful:
“It’s so slow.”

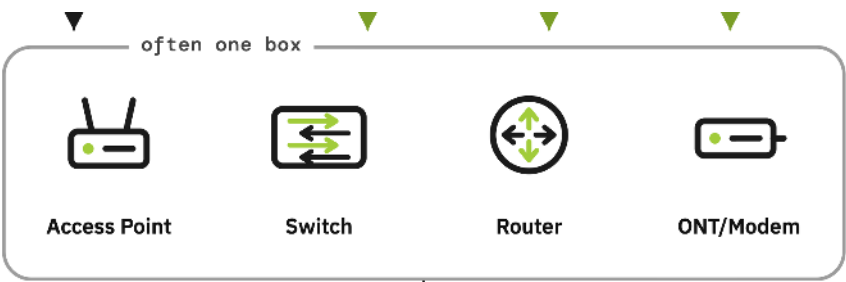


More useful:
Wi-Fi: Strong / Internet: Weak / here are the readings

Measurements give support somewhere useful to begin.

From feeling to evidence. The first is what support hears all day, and there is nothing to work with in it. The second is a set of readings copied straight off your device, in the app’s own words, which gives the person helping you somewhere useful to begin. Same problem, completely different conversation.

That is the payoff of every reading you learned to run in this book. Not so you could become an engineer. So you could walk into the conversation holding something you measured instead of something you felt.



Four jobs, one box, zoomed out of the big picture. In many homes several of these jobs are combined into one box on the wall: handing out Wi-Fi, connecting the wires, routing the traffic, and holding the link to your provider. **You do not need these words to fix anything.** You need them for one thing only, which is this chapter: when somebody on a support line asks which part you mean, you can point at the right one instead of saying “the internet box.”

What to actually say

Keep it short. The fact does the work, so you do not have to. Here is the shape of it, for any of the handoff doors from the last chapter.

For the provider, when the internet feed read Weak: “I ran a connection test. My Wi-Fi reads Strong, my internet feed reads Weak. The problem is the feed coming into the house. I’ve pasted the result below.” Then paste it.

For the installer or office IT, when your home fixes have run out: “Here’s what I’m seeing.” Then paste the reading. Let them take it from there. You are not telling them how to do their job. You are handing them a clean starting point so they do not waste an hour rediscovering what your device already knows.

And if they say a word you don’t know, you already have the Glossary from the last chapter. Look it up, right there, and stay in the conversation as an equal.

The whole script is: state what you measured, paste the evidence, stop talking. You do not need to argue. You do not need to know more than you know. **The reading is worth something because it is a thing you measured rather than a thing you guessed**, and if you ran it twice, on two devices, it is worth a good deal more than that.

Do this now

One run, one copy, one paste. This is the habit that makes you credible the next time something breaks.

1. **Run Test My Connection and copy the result.** It does not matter what it says today. The point is to find the copy action with your own thumb and feel how easy it is.

2. **Paste it into a message to yourself.** Open a note or a text to your own number and paste the reading in. Read it back. That is exactly what a provider or installer would see, and notice how much more solid it looks than anything you would have typed in frustration.
3. **Save that habit for the real day.** The next time something genuinely breaks and it lands on the wrong side of the do-it-yourself line, you will not reach for angry words. You will reach for the copy action, paste the fact, and send it. You will be a much easier caller to help.

Do it once now, while nothing is broken, so the move is already in your hands when something is.

Myth-Bust: “All I can tell them is that it’s slow.”

No. You can tell them exactly what you measured, and that is a completely different conversation. The reason people feel unheard on these calls is that they arrive with a feeling, and there is very little anyone can do with a feeling except start at step one. “My Wi-Fi reads Strong and my internet reads Weak” is worth more than “it’s terrible,” and it is worth more again when you ran it twice. What changes the call is a person who pastes in three measured readings and says “my Wi-Fi is Strong, my feed is Weak, here’s the result.” That person is not guessing, and the help desk knows it. You went from the least credible kind of caller to the most credible, and all it took was one tap to copy and one to paste. The facts were on your device the whole time. Now you know how to hand them over.

The end of the book, and the one page to keep

Here is where we are. You started this book able to say four words, “my Wi-Fi is broken,” and not much else. Look at what you can do now.

You can tell, in about a minute, where to start looking: your Wi-Fi, the internet side, or the app itself. You can read the handful of plain facts your own device keeps about its connection. You can tell apart the different kinds of slow, so you chase the right fix instead of the loudest one. You can make one change at a time and watch the grade move, so you find out whether a fix actually worked instead of hoping it did. You know when the evidence says to stop changing your Wi-Fi and hand the problem to someone else, and you know who that is. And now you know how to hand it to them so you are taken seriously the first time.

That is not a small thing. That puts you ahead of nearly everyone who has ever stood in a back bedroom, holding a phone in the air, blaming the wrong

road.

I have spent over twenty-five years learning that most Wi-Fi problems are not Wi-Fi problems. You learned it in an afternoon, with a free app and your own two thumbs. The radio will keep doing its quiet work whether you understand it or not. What you have now is the ability to look, instead of guess, and that is the whole game.

There is one page left that matters, and it is the one to keep. Tear out the Quick-Reference Card, or save it to your device, and put it where the box lives. It holds the one-tap check, the four answers and what each means, the kinds of slow, the home fixes in order, and the one line that tells you to stop and call someone. It is the entire book on a single page, sitting right where you will need it. Everything deeper lives in the app, ready whenever you tap it.

The next time someone in your home says “the Wi-Fi is broken,” you will not reach for the box. You will reach for your device, run the check, read the answer, and know exactly what to do. Welcome to the short list of people who actually do.

Quick-Reference Card

Keep this one page by the box on the wall.

Fix Your Own Wi-Fi: the one page

1 START HERE

run Test My Connection



2 WHERE DO I LOOK FIRST

Wi-Fi side weaker → look at the local connection

Internet side weaker → run it again, then work outward

Both healthy → try another app before changing anything

Could not check → run it again rather than guess



3 IF IT REALLY IS WI-FI

- 1** Reconnect the device
- 2** Move closer
- 3** Check the network
- 4** Get the box up and out
- 5** Reboot the right box, once



4 THE STOP LINE

copy the result, hand it off

Appendix: The WLAN Pros Toolbox

Everything this book asks you to do happens in one free app. This page is here so you never have to go looking for it, and so you can hand it to someone else.

Getting it

The one address to remember is toolbox.wlanpros.com. Open it on the device you want to test and it sends you to the right place for whatever you are holding. If you are reading this on a screen, that link is tappable. If you are holding paper, the address is short enough to type.

It is free. There is no account to make, nothing to buy, no ads, and nothing is collected about you or your network.

Your device	Where it comes from
iPhone or iPad	The App Store
Android phone or tablet	Google Play
Mac	A download from toolbox.wlanpros.com
Windows	The Microsoft Store
Anything with a browser	toolbox.wlanpros.com/app

You do not need to pick the right one from that list. Go to toolbox.wlanpros.com on the device you want to test and it works the rest out for you.

The four screens this book uses

Everything in these seven chapters runs on four screens. If you learn nothing else in the app, learn these.

Test My Connection. The one-tap front door. Two grades, one for your Wi-Fi and one for your internet, because there are two roads. This is Chapter 1.

Wi-Fi vs Internet. Puts the two roads side by side and names which one is the tighter of the two right now. This is Chapter 2, and it is the screen that

saves you money.

Wi-Fi Information. The five facts about your own connection: which network, which band, which channel, how strongly you hear the box, and the rate you agreed to try at. This is Chapter 3.

Network Quality. Six readings instead of one big number, because slow is four different experiences. This is Chapter 4.

What else is in there

You bought a book about fixing your Wi-Fi, and the app you installed for it happens to be the field toolbox a lot of professionals carry. You are welcome to the rest of it.

- **Calculators.** Signal math, path loss, EIRP, link budget, throughput, power over Ethernet budgets, subnet and IPv6 math.
- **Reference tables.** Standards, channel charts, signal thresholds, security types, power classes.
- **A searchable glossary.** Every term in this book and several hundred more, offline, so it works when your connection does not. This is the screen Chapter 6 sends you to when somebody uses a word you do not know.
- **Printable reference cards.** The laminated cards WLAN Pros makes, as PDFs you can carry or print.
- **Network diagnostics.** Ping, DNS lookup, traceroute, and the rest of the tools you would find on an engineer's laptop.

None of it is required reading. It is there the day you get curious.

If something is wrong

Email support@wlanpros.com. And if you reach the point where you need to describe a problem to anyone, use the copy action in the app and paste the readings. That is the whole of Chapter 7.

About the Author

Keith Parsons is CWNE #3, one of the first in the world to earn the Certified Wireless Network Expert credential, and the founder and host of the Wireless LAN Professionals Conference (#WLPC). For twenty-five years he has worked in nothing but wireless, designing Wi-Fi in 117 countries and across nearly every industry on earth, from a diamond mine 800 meters underground to the top of the Empire State Building. Along the way he has taught over ten thousand people how to design and troubleshoot Wi-Fi.

The WLAN Pros Toolbox, the free app used throughout this book, is his answer to the same question he has heard for a quarter century: “My Wi-Fi is broken.” Now you can answer it yourself.